UM-2: a 4+4 model of the Universe

Einstein-Lovelock gravity, the dirac16complex field, three generations of quarks and leptons with three neutrino flavours, their mixing and their flavour conversion, photons, gluons, and the W^+, W^- and Z bosons without a Higgs particle; every mass from the x_0 dependence of wave functions

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Model definition, version 3, 2026-10-18 (version 2: 2026-10-18; version 1: 2026-10-17)

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1 How to read this document

This document DEFINES the model UM-2. Every field, every interaction and every equation of the model is written out here; nothing has to be looked up elsewhere. Each statement carries one of these labels:

- **DERIVED**: computed and verified in this work (sympy or numerics; the verification is stated with the result);
- **AUTHOR**: taken from the author's notebooks (coordinates, metric, Lovelock weights, gamma matrices, mass parameter);
- **STANDARD**: textbook physics used as an input;
- **ASSUMED**: a choice made for this model;
- **INVENTED**: a new phenomenological ingredient of this model;
- **ESTIMATE**: an order-of-magnitude evaluation, not a precision computation;
- **OPEN**: not resolved in this version.

"(record)" after ASSUMED marks a choice taken over from the Revision record of this repository. Units: $\hbar = c = 1$ unless stated. Indices $M, N, \ldots$ run over the eight coordinates, $a, b, \ldots$ over the orthonormal frame, μ, ν over the four observable coordinates x_1, x_2, x_3, x_4.

Version 2 (2026-10-18). Version 1 contained one Standard-Model generation and used a band of dirac16complex as a stand-in for the missing neutrinos ν_μ, ν_τ. Version 2 contains three full generations of quarks and leptons, anomaly free generation by generation, with three neutrino flavours, their masses, their mixing (the PMNS matrix) and their flavour conversion in vacuum, in matter, in the Sun, in the early Universe and in supernovae; the quark mixing (the CKM matrix) follows from the same mechanism. The chiral tip Z portal of version 1 is removed (section 5.7). Every other part of the model is unchanged.

Version 3 (2026-10-18). By the author's order of 2026-10-18 to find and document an accurate approximate

solution of the coupled field equations, version 3 adds: a third portal, I13, an x_0-chiral vector coupling of dirac16complex to the baryon-number current, which gives the dark matter its relic abundance by s-wave annihilation (the cross-section respects the $J = 1$ unitarity bound, which it reaches only in the collisions that carry 22 % of the freeze-out rate; section 5.7); definite benchmark strengths of the portals I11 (with the $B - L$ charges; it produces the fold bands of dirac16complex by freeze-in, $\Delta N_{eff} = 0.001$) and I12 (universal coefficients, unitarity safe); the coupled solution of the Kohn-Sham equations of dirac16complex together with gravity and every Standard-Model field in ten environments, from the cosmic mean density to neutron-star cores and the early Universe (section 9.1); the thermal history of the model with its x_0 masses and the dark-matter relic from the Boltzmann equation (section 11.2); the magnitudes of the terms of the metric power series and σ_8 (sections 4.3 and 12); and the Majorana alternative to the Dirac neutrinos (section 10.8). Version 2 found that I12 alone cannot produce the relic abundance within perturbative unitarity; version 3 keeps that result and resolves it with I13. An independent review of version 3 led to these corrections, which this text contains: I13 is summed over the seven directions $M \neq x_0$ (for $M = x_0$ its bilinear is not Hermitian); the block responds to the hidden pressure of the hot plasma, not only to that of cold matter (section 9.1); the self-interaction λ of Ψ would let the dark matter thermalise the fold bands, so the cosmology requires a small λ (section 11.2); and the cosmological results are those of the model with the freeze-in radiation (section 12).

Symbols with more than one use:

- k is the block variable $1/f^2$ (k_*, $k_\pm$, δk, $G(k)$); as the index of α_k, $E_{(k)}$ and $\mathcal{L}_{(k)}$ it is the Lovelock order (written p in the reduction formula of section 8.5, where the block variable appears as well); in e^{ikx_1}, in h_j, in $\vec{k}$, in Δk and in the dispersion $\varepsilon = e^{-a_4}|k|$ it is the momentum of a mode along x_1, x_2, x_3.
- $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue (Λ_n, Λ_W, Λ_Z, Λ_D are particular ones); $\Lambda_\star$ is the scale of the effective interactions; Λ_ℓ and Λ_q are the tip coupling matrices.
- n is a number density (n_o that of a Kohn-Sham orbital), except as a level or period number (Λ_n, $z \to n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell.
- The subscript t (x_t, p_t, v_t) refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate.
- f is the block scale and f_o an occupation number. ρ is an energy density, except for the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$; ϱ is a probability density in ξ.
- $w = (w_1, w_2, w_3)$ are the Lovelock weights; w is also the weight 1/2 of a fold parity in the Kohn-Sham sums, $\sqrt{c^2 - e^2}$ in the fermion mass condition, and the equation-of-state parameter p_3/ρ. $z = 6Hx_0$, except where it is a redshift (dark energy, cosmology).
- Flavour: U is the PMNS matrix and V the CKM matrix (the gauge transformations are U_L, U_Y, the block basis V_β, the transverse volume $V_\perp$, and the matter potentials always carry a subscript, V_{CC}, V_{NC}); κ_ν, κ_e, κ_u, κ_d are the 3×3 tip matrices of the four fermion sectors (κ_f the tip mixing of one generation, $\kappa = 8\pi G_8$ without subscript); θ_{12}, θ_{13}, θ_{23}, δ are mixing angles and the Dirac phase (θ_W the weak angle, θ also the parameter of a gauge transformation; δ also a variation, as in δk), $s_{ij} = \sin\theta_{ij}$ and $c_{ij} = \cos\theta_{ij}$; J is the Jarlskog invariant, except in the block operator $J = \gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$.
- Generations and flavours: $i = 1, 2, 3$ is the generation index of the fermion fields and labels the neutrino mass states (in the curvature formulas i, j are 3-space indices); $\alpha \in \{e, \mu, \tau\}$ is a flavour index (on G^α and T^α a colour index).
- Fermion names: e, μ, τ are the charged leptons, ν_1, ν_2, ν_3 the neutrino mass states and ν_e, ν_μ, ν_τ the flavour states; u, c, t and d, s, b the quarks. The same letters have other meanings: e is also the elementary charge and the dimensionless fermion eigenvalue εfL; μ_ψ is a bulk mass and $\mu = \sqrt{2}G_F n_{\nu_e}$ the unit of the neutrino self-interaction; c with a subscript (c_L, c_Q, c_R, c_ψ) is a localisation $f\mu L$; s is also $\sinh w/w$ in the fermion mass condition; t as a subscript refers to the extra times.
- H is the author's constant (an inverse length), except in the neutrino-decoupling equations, where $H = \sqrt{8\pi\rho/3}/M_{Pl}$ is the Hubble rate, $M_{Pl} = G_N^{-1/2} = 1.22091 \times 10^{19}$ GeV the Planck mass and ρ the total energy density. The time is always x_4, and the symmetry axis of the fast-flavour analysis is x_3. Λ^0, Λ^3 (upper indices) are the potentials of the fast-flavour analysis, $\Lambda_0 = -24H^2$ the cosmological constant of the sourceless block. $G(k)$ is the Einstein-Hilbert coefficient and $G(u)$ the angular distribution of the electron lepton number; $F_h = 1.0095$ is the heating factor of ΔN_{eff} and $F(T_1, T_2)$ the energy-transfer kernel of neutrino decoupling.
- Portals (version 3): G_V (I11, GeV^{-2}), G_g, G_W, G_B and their combination $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2}$ (I12, GeV^{-3}) and G_A (I13, GeV^{-2}) are four-dimensional coupling strengths (G_N and G_F are Newton's and Fermi's constants, $G(k)$ the Einstein-Hilbert coefficient); $\tilde{I}_{of}$ and $\tilde{I}_{of}^{(b)}$ are overlap integrals of a dirac16complex orbital

o with a Standard-Model zero mode f; q_f is a $B-L$ charge; $P_a=(1+\gamma^{x_0})/2$ and $P_b=(1-\gamma^{x_0})/2$ are the x_0-chiral projectors (not the four-dimensional chiral projector $P_+=(1+\Gamma_4)/2$); J^B_M is the baryon-number current.

- Thermal history (version 3): $x=M/T$ with $M=M_{DM}$ the dark-matter mass (not the author's M of $m=3HM$; M and m are also the masses of two attracting bodies in section 4.3); $Y=n/s$ is an abundance (s the entropy density, not $\sinh w/w$, and in the cross-sections s is the squared centre-of-mass energy, the Mandelstam variable); g_ρ, g_s (and g_*) are effective numbers of degrees of freedom; γ is a rate density (pairs per volume and time) and, in the supernova analysis, the adiabaticity parameter; σ is a cross-section, the radion σ of the metric series and σ_8 the amplitude of the matter fluctuations; K_1, K_2 are modified Bessel functions (the gamma products K_1, K_2 of section 6.2 carry no argument); J is a total angular momentum in the partial-wave bounds; $\beta=\sqrt{1-4M^2/s}$ is the centre-of-mass velocity of each annihilating particle (the tip coefficients β_W, β_Z carry a subscript); $\delta=\frac{1}{3}\,d\ln g_s/d\ln T$ in the Boltzmann equation; f is also the logarithmic growth rate in $f\sigma_8$; a is the cosmological scale factor and, in the graviton profiles, $a=3\eta/2$; b is the QCD beta-function coefficient $11-2n_f/3$ (b_2, b_Y for $SU(2)$ and $U(1)_Y$) and b, c in the thermal average are the coefficients of $\langle\sigma v\rangle=a+6b/x$; N_c is the colour multiplicity of a Weyl component, $N_\chi=2$ the number of fold bands (of Ψ_A and Ψ_B) and $N_F=2$ the number of final fold species; ϵ is the order of the metric series and Δ_Φ the rms Newtonian potential per logarithmic interval of the wavenumber; α_n (with a level index) is the Yukawa strength of a Kaluza-Klein graviton; G_λ and $\tilde{I}_\lambda$ are the four-dimensional strength and the overlap of the self-interaction channel (section 11.2), $\lambda_B=7\times10^{-5}H^{-6}$ the self-coupling of the numerical Kohn-Sham tests; p_2 is the order-α pressure of a gauge plasma and $\bar{p}=(p_0+3p_t)/4$ the hidden pressure.

2 Summary

1. **Spacetime.** Eight dimensions with the author's coordinates $X=\{x_0,x_1,x_2,x_3,x_4,x_5,x_6,x_7\}$: x_0 is the hidden space-like direction, x_1,x_2,x_3 are ordinary space, x_4 is the time, and x_5,x_6,x_7 are three extra time-like directions. The frame metric is $\eta=\mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ in this order. The metric depends on x_0 only through $z=6Hx_0$ with period π in z, i.e. period $\ell_0=\pi/(6H)$ in x_0.
2. **Gravity.** Einstein-Lovelock gravity in eight dimensions with the author's weights $w=(1,1/32,1/1152)$ and $\Lambda=-10H^2$, kept unchanged. A block stabiliser (a vacuum energy of the hidden block, ASSUMED) gives a stable static block at $k_*=12+\frac{4}{3}\sqrt{51}$ on which four-dimensional gravity is attractive: its Einstein-Hilbert coefficient is $G(k_*)=\frac{20}{3}+\sqrt{51}=13.81>0$ (DERIVED).
3. **The deflating extra times.** In the primordial era the extra times x_5,x_6,x_7 (and x_0 with them) deflate at rates up to $2.6H$ into the stabilised block, then the block oscillates with the radion frequency $3.387H$ while 3-space keeps expanding (DERIVED, full nonlinear equations).
4. **dirac16complex** Ψ: a 16-component complex anticommuting spinor of $Cl(4,4)$ (Pin(4,4) irreducible; two inequivalent 8-dimensional irreducible representations of Spin(4,4)) coupled to gravity through the canonical spin connection, which is non-trivial: $\gamma^M\Omega_M=\frac{3}{2}(H/f)\gamma^{x_0}+(\dot{b}_4/b_4-\frac{1}{2}\dot{a}_4)\gamma^{x_4}$. It splits into two Lorentz-invariant halves, Ψ_A ($B=-1$) and Ψ_B ($B=+1$), with $B=-iC\gamma^{x_4}$ the Krein metric.
5. **Standard-Model content (three generations, no Higgs particle).** Photon, gluons, W^+ (charge $+e$), W^- (charge $-e$), Z (neutral), and three generations of quarks, (u,d), (c,s), (t,b), and of leptons, (ν_e,e), (ν_μ,μ), (ν_τ,τ), each generation with a right-handed neutrino. The gauge anomalies cancel generation by generation. Electroweak symmetry is broken at the regular tip of the hidden direction by a non-linear sigma field $\Sigma\in SU(2)$ that has no radial mode, hence no Higgs particle.
6. **Masses and mixing.** Every four-dimensional mass is the eigenvalue of an x_0 operator acting on a wave function periodic in x_0 with period $\pi/(6H)$. Massless: graviton, photon, gluons (flat x_0 profiles) and the fold band of Ψ (chiral zero modes). Massive: $W^\pm$, Z (profiles bent at the tip), the twelve fermions (a doublet zero mode and a singlet zero mode of each fermion mixed at the tip by a 3×3 tip matrix in each sector), the bulk band of Ψ and every Kaluza-Klein mode (DERIVED spectra, fitted to the measured masses with order-one parameters). The doublet bulk mass is common to the three generations of a sector; then the left-handed mixing matrices are $U=\kappa_e^\dagger\kappa_\nu$ and $V=\kappa_u^\dagger\kappa_d$ (DERIVED, exact); with $\kappa_e=\kappa_u=1$ (ASSUMED) they are the tip matrices κ_ν and κ_d, whose values are the measured PMNS and CKM matrices (STANDARD inputs).
7. **Three neutrino flavours.** Dirac neutrinos with masses 1.000, 8.712 and 50.140 meV (normal ordering, the NuFIT 6.0 mass splittings, the lightest mass ASSUMED) and the PMNS mixing of NuFIT 6.0 (Dirac phase 212°). Their flavour conversion is computed in vacuum, in matter (the MSW effect), in the Sun, at accelerators, in the early Universe (where it accompanies neutrino decoupling, $N_{eff}=3.0432$), in the cosmic

neutrino background (three mass eigenstates) and in supernovae (the MSW resonances of the envelope, the fast flavour instability of the dense neutrino gas and its effect on the explosions).

8. **New interactions (INVENTED).** Three portals couple Ψ to the Standard Model: a gauge-kinetic portal I12 couples $\bar{\Psi}\Psi$ to pairs of gauge bosons (W^+W^-, ZZ, $Z\gamma$, $\gamma\gamma$, gluon pairs; benchmark strength $(38.2\ \text{TeV})^{-3}$, unitarity safe); a weak vector contact portal I11 couples the whole field to the $B-L$ fermion currents (benchmark $G_V = 4.4\times10^{-17}\ \text{GeV}^{-2}$: it produces the massless fold bands by freeze-in, $\Delta N_{eff} = 0.001$); and (version 3) an x_0-chiral vector portal I13 couples the current $i\bar{\Psi}\gamma^M P_b \Psi$ ($M \neq x_0$), which vanishes on the fold bands, to the baryon-number current ($G_A = 1.28\times10^{-9}\ \text{GeV}^{-2}$). The chiral tip Z portal of version 1 is removed: with three neutrinos it has no role, it is bounded by the Z width and disfavoured by N_{eff} (section 5.7).
9. **The dark sector.** The lightest odd states of the two halves Ψ_A and Ψ_B, degenerate Dirac fermions of mass $3.1022\,m = 44.35$ TeV ($m = H$), are electrically and weakly neutral, stable by the fold parity, cold, and together the dark matter. I12 alone cannot produce their relic abundance within perturbative unitarity (version 2); with I13 they annihilate in the s-wave into quark pairs and freeze out with $\Omega h^2 = 0.060$ each, 0.120 together (Boltzmann equation with the thermal history of the model; DERIVED, section 11); their scattering on xenon nuclei, $4.0\times10^{-46}\ \text{cm}^2$ per nucleon, is 0.16 of the present LZ limit. The self-interaction of Ψ must be weak, $G_\lambda \leq 2.3\times10^{-15}\ \text{GeV}^{-2}$ (ASSUMED), or the dark matter would thermalise the fold bands (section 11.2). No state of Ψ is dark energy; the dark energy of UM-2 is a tuned four-dimensional cosmological term.
10. **Cosmology.** The four-dimensional effective theory of the stabilised block is exact general relativity, so the CMB, the CNB and the BAO follow from the standard Einstein-Boltzmann equations with the UM-2 particle content; $\sigma_8 = 0.812$. The metric is a power series of terms of decreasing magnitude: the order-one cosmological potentials are 2.4×10^{-5} and smaller, the order-two terms 6×10^{-10}, the radion 10^{-34}, and the Kaluza-Klein gravitons act only below 10^{-21} m.
11. **The coupled solution (version 3).** The Kohn-Sham equations of dirac16complex, coupled to gravity (the block response) and to the mean fields of every Standard-Model field (I11, I12, I13), are solved self-consistently from the cosmic mean density to neutron-star cores and through the early Universe: the back-reactions are small (the dark-matter level moves by at most 1.5×10^{-6} of its mass, at the dark-matter freeze-out; the block by $\delta\ln f \leq 1.1\times10^{-34}$, pushed by the hot plasma at the freeze-out, and by at most 10^{-48} from nucleosynthesis on), so the solution is the stabilised background with the x_0 spectra, the Kohn-Sham states and the linear cosmological perturbations, and the coupled corrections are computed (section 9.1).

3 Space, coordinates and the periodic hidden direction

3.1 The coordinates (AUTHOR)

The author's coordinates are $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ (the author's notebook: `X = {x0[], x1[], ..., x7[]}`), with

- x_0: the hidden space-like direction, $z \equiv 6Hx_0$ takes every real value;
- x_1, x_2, x_3: ordinary 3-space;
- x_4: the time;
- x_5, x_6, x_7: three extra time-like directions.

$H > 0$ is the author's constant. Frame indices $a = 0, \ldots, 7$ are aligned with $x_0, \ldots, x_7$, and

$$\eta_{ab} = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1).$$

3.2 The hidden coordinate y, the fold and the tips (DERIVED)

On every interval of z on which $\sin z \neq 0$ define

$$y = \frac{\ln|\sin z|}{6H} \in (-\infty, 0], \qquad |\sin z| = e^{6Hy}, \qquad dy = \cot z\, dx_0.$$

One period $0 < z < \pi$ of x_0 (length $\ell_0 = \pi/(6H)$) consists of two halves, $0 < z < \pi/2$ and $\pi/2 < z < \pi$, joined at the **fold** $z = \pi/2$ ($y = 0$, where $\cos z = 0$). Each half ends at a **tip** ($z \to 0^+$ or $z \to \pi^-$, $y \to -\infty$), at infinite

proper distance from the fold. As in the Kohn-Sham record of this repository the tip is replaced by a regular cutoff at $y = -L$ (ASSUMED), with

$$\eta \equiv HL = 3.$$

The fold reflection

$$R: \quad z \mapsto \pi - z \qquad (x_0 \mapsto \ell_0 - x_0)$$

maps the two halves onto each other; modes of a field on one period are even or odd under R. Every function of $y(z)$ is periodic in x_0 with period $\ell_0 = \pi/(6H)$.

4 The metric as a power series

The metric is a series of terms of decreasing magnitude,

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2 h^{(2)}_{MN} + \ldots, \qquad \epsilon \sim 10^{-5},$$

where ϵ is the size of the CMB temperature anisotropy (STANDARD).

4.1 Order 0: the corrected primordial metric (AUTHOR coordinates, DERIVED properties)

In the author's chart, with two general functions of the time $a_4(x_4)$ and $b_4(x_4) > 0$,

$$\bar{g} = \mathrm{diag}\Big(e^{-2a_4} b_4 \cot^2 z,\ e^{2a_4},\ e^{2a_4},\ e^{2a_4},\ -1,\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -e^{-2a_4} b_4 (\sin^2 z)^{1/6}\Big),$$

where $(\sin^2 z)^{1/6}$ is the positive branch. This is the metric of the Revision record of 2026-10-16 (version 2), whose sources are all diagonal. The notebook metric

$$g^{\mathrm{nb}} = \mathrm{diag}\big(\cot^2 z,\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ e^{2a_4}(\sin^2 z)^{1/6},\ -1,\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6},\ -e^{-2a_4}(\sin^2 z)^{1/6}\big),$$

to whose extra-time components the author's order of 2026-10-14 added the factor b_4, differs from it in g_{00} and in the 3-space warp. Define

$$f = e^{-a_4} b_4^{1/2}, \qquad k = \frac{1}{f^2} = \frac{e^{2a_4}}{b_4}, \qquad A = \dot{a}_4, \qquad h = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4,$$

dots being d/dx_4. In the y chart the metric is

$$d\bar{s}^2 = e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 + f^2\big[dy^2 - e^{2Hy}(dx_5^2 + dx_6^2 + dx_7^2)\big].$$

The block $\{x_0, x_5, x_6, x_7\}$ is f^2 times a four-dimensional space of constant curvature $-H^2$ and signature $(1,3)$; f is its common scale factor ("the block"), A is the observer's Hubble rate and $h < 0$ means that the extra times and x_0 deflate. The volume element and the vielbein are

$$\sqrt{|\bar{g}|} = e^{3a_4} f^4\, |\cot z|\, (\sin^2 z)^{1/4} = e^{3a_4} f^4 e^{3Hy} \left|\frac{dy}{dx_0}\right|,$$

$$e^0{}_{x_0} = f \cot z, \quad e^i{}_{x_i} = e^{a_4}\ (i = 1,2,3), \quad e^4{}_{x_4} = 1, \quad e^t{}_{x_t} = f(\sin^2 z)^{1/12}\ (t = 5,6,7).$$

Curvature (DERIVED; sympy, identical in both charts, 25/25 checks). The Riemann tensor is pair-diagonal, $R^{ab}{}_{cd} = 0$ unless $\{a,b\} = \{c,d\}$, and every sectional curvature $K_{ab} = R^{ab}{}_{ab}$ is one of

$$K_{ij} = A^2, \quad K_{i4} = A^2 + \dot{A}, \quad K_{iB} = Ah, \quad K_{4B} = h^2 + \dot{h}, \quad K_{BB'} = h^2 - H^2 k,$$

with $i, j \in \{1,2,3\}$ and $B, B' \in \{0,5,6,7\}$. None depends on x_0; the curvature is the same on both halves of every period.

4.2 Order 1: the observable perturbations and the hidden tower

Every field and every metric perturbation is expanded in the eigenfunctions $u_n(x_0)$ of its x_0 operator (section 6),

$$h^{(1)}_{MN}(x^\mu, x_0) = \sum_{n\geq 0} h^{(1,n)}_{MN}(x^\mu)\, u_n(x_0).$$

The $n = 0$ term has the flat profile $u_0 = \text{const}$; in the Newtonian gauge its line element is

$$ds^2 = e^{2a_4}\big[(1 - 2\epsilon\Psi_N)\delta_{ij} + \epsilon h^{TT}_{ij}\big]dx_i dx_j - (1 + 2\epsilon\Phi)dx_4^2 + f^2(1 + 2\epsilon\sigma)\big[dy^2 - e^{2Hy}dx_t^2\big],$$

with the two scalar potentials Φ, Ψ_N, the transverse-traceless tensor h^{TT}_{ij} and the radion σ (the relative fluctuation of the block size). The scalar and tensor sectors carry the CMB, the CNB and the BAO; the radion is massive (section 8.3) and does not propagate at cosmological distances. The terms $n \geq 1$ are Kaluza-Klein modes with masses above the gap $3H/(2f)$ (section 6). Order 2 contains the products of order-1 terms and the back-reaction of x_0-localised sources; its magnitude is estimated below, its detailed form is not computed (OPEN).

4.3 The magnitudes of the terms (DERIVED, version 3; the order-2 sizes are ESTIMATES and the tensor bound is STANDARD)

The solution of the coupled equations fixes the size of every term (4 checks):

- **Order 0**, $\bar{g}$: magnitude 1.
- **Order 1,** $n = 0$ (the Newtonian potentials). On cosmological scales the rms potential per logarithmic interval of the comoving wavenumber k today, from the Weyl potential $\frac{1}{2}(\phi + \psi)$ of the linear solution (section 12),

  $$\Delta_\Phi(k) = \Big[\frac{k^3 P_\Phi(k)}{2\pi^2}\Big]^{1/2}$$

  (P_Φ the power spectrum of the potential) is 2.42×10^{-5}, 2.28×10^{-5}, 1.51×10^{-5}, 4.12×10^{-6}, 1.74×10^{-6}, 3.55×10^{-7} and 5.03×10^{-8} at $k = 10^{-4}$, 10^{-3}, 10^{-2}, 0.05, 0.1, 0.3 and 1 Mpc^{-1}. It agrees with the Poisson equation $\Phi = \frac{3}{2}\Omega_m H_0^2\, \delta_m/(ak^2)$ ($\Omega_m = 0.3150$, $H_0 = 67.38$ km/s/Mpc, δ_m the matter density contrast, a the scale factor) applied to the linear matter density to 1.5×10^{-6} for $k \geq 10^{-2}$ Mpc^{-1}, and on the largest scales with $\frac{3}{5}A_s^{1/2}(k/k_0)^{(n_s-1)/2}g(0)$ ($k_0 = 0.05$ Mpc^{-1}) to 4.4×10^{-5}, where $g(0) = D(1) = 0.7878$ is the growth suppression by Λ_4, $D(a) = \frac{5}{2}\Omega_m E(a)\int_0^a db/[bE(b)]^3$, $E(a) = (\Omega_m a^{-3} + 1 - \Omega_m)^{1/2}$. Bound objects have $\Phi = G_N M/(Rc^2)$: the Earth's surface 7.0×10^{-10}, the Sun's 2.1×10^{-6}, a white dwarf (0.6 solar masses, 8700 km) 1.0×10^{-4}, a neutron star (1.4 solar masses, 12 km) 0.17; galaxies and clusters $\Phi \sim (v/c)^2 = 5.4 \times 10^{-7}$ ($v = 220$ km/s) and 1.1×10^{-5} (1000 km/s). The series converges quickly everywhere except near neutron stars and black holes, where the full nonlinear equations apply.
- **Order 1,** $n \geq 1$ (the Kaluza-Klein gravitons). The graviton has no tip term ($\beta = 0$), so its even levels are $q_n = n\pi$, $\Lambda_n = 9\eta^2/4 + n^2\pi^2 = 30.1196$, 59.7284, 109.0764 ($n = 1, 2, 3$; finite elements agree to 3.9×10^{-9}), with

  $$u_n(\xi) = e^{-a\xi}\Big(\cos q_n\xi + \frac{a}{q_n}\sin q_n\xi\Big), \qquad a = \frac{3\eta}{2}, \qquad u_n(0) = 1,$$

  $$N_n = \int_{-1}^{0} e^{3\eta\xi}u_n^2\, d\xi = \frac{1}{2}\Big(1 + \frac{a^2}{q_n^2}\Big), \qquad N_0 = \frac{1 - e^{-3\eta}}{3\eta}.$$

  Two static masses M and m with the normalised x_0 profiles ϱ_M and ϱ_m attract at the distance r with

  $$V(r) = -\frac{G_N M m}{r}\Big[1 + \sum_{n\geq 1}\alpha_n e^{-m_n r}\Big], \qquad \alpha_n = \frac{N_0}{N_n}\Big(\int \varrho_M\, u_n\, d\xi\Big)\Big(\int \varrho_m\, u_n\, d\xi\Big), \qquad m_n = \frac{\sqrt{\Lambda_n}}{fL};$$

  for two masses with the same profile ϱ the strength is $(N_0/N_n)(\int \varrho\, u_n\, d\xi)^2$: $\alpha_n = 0.0728$, 0.1469, 0.1809 for two sources at the fold, 0.0904, 0.0136, 0.0057 for the quark-doublet zero-mode profile and 0.0208, 0.0197, 0.0105 for the lepton doublet; a source with the flat profile of the gauge zero modes (the gluon field energy, which makes up most of the nucleon mass) excites none ($\int e^{3\eta\xi}u_n\, d\xi = 0$, orthogonality to u_0). $m_1 = m_{KK} = 121.32$ TeV, whose range $\hbar c/m_1 = 1.63 \times 10^{-21}$ m makes $e^{-m_1 r} = e^{-3.1\times 10^{16}}$ at 50 micrometres: Newton's law is exact at every measurable distance.

- **The radion σ**: its static response to matter is at most $\delta \ln f = 1.1 \times 10^{-34}$ (at the dark-matter freeze-out, from the hidden pressure of the hot plasma; 10^{-62} for the Earth; section 9.1); its range is 4.08×10^{-21} m; the primordial oscillation decays (lifetime $0.43\,\mathrm{s}/C_r$, section 14).
- **Order 2** (ESTIMATE: the squares of the first-order terms): 5.8×10^{-10} on the largest cosmological scales (Δ^2_Φ at $k = 10^{-4}$ Mpc^{-1}), 4.5×10^{-12} at the Sun, 0.03 at a neutron star.
- **Tensor modes h^{TT}_{ij}**: their primordial amplitude is not predicted (the primordial spectrum is an input of UM-2); the bound $r < 0.036$ on the tensor-to-scalar ratio (STANDARD, from memory) gives $\Delta_h < (rA_s)^{1/2} = 8.7 \times 10^{-6}$.

On cosmological scales the terms therefore decrease as 1, 2.4×10^{-5} (order 1), 5.8×10^{-10} (order 2), 10^{-34} (radion) and 0 (Kaluza-Klein gravitons at any measurable distance): the $n = 0$ term of order 1 carries the CMB, the CNB and the BAO.

5 The action

$$S = S_{EL}[g] + S_{stab}[g] + S_\Psi[g,\Psi] + S_{gauge}[g,G,W,B] + S_f[g,\psi;G,W,B] \\ + S_{tip}[g;W,B,\Sigma,\psi] + S_{portal}[\Psi;\psi,G,W,B,\Sigma].$$

5.1 Einstein-Lovelock gravity (AUTHOR weights)

$$S_{EL} = \frac{1}{2\kappa}\int d^8x\,\sqrt{|g|}\,\Big(\sum_{k=1}^{3}\alpha_k\,\mathcal{L}_{(k)} - 2\Lambda\Big), \qquad \mathcal{L}_{(k)} = \frac{1}{2^k}\,\delta^{a_1b_1\cdots a_kb_k}_{c_1d_1\cdots c_kd_k}\,R^{c_1d_1}{}_{a_1b_1}\cdots R^{c_kd_k}{}_{a_kb_k},$$

with the generalised Kronecker delta $\delta^{\cdots}_{\cdots}$ (the determinant of ordinary deltas). $\mathcal{L}_{(1)} = R$, $\mathcal{L}_{(2)} = R^2 - 4R_{ab}R^{ab} + R_{abcd}R^{abcd}$ (Gauss-Bonnet) and $\mathcal{L}_{(3)}$ is the cubic Lovelock density. The field equations are

$$\sum_{k=1}^{3}\alpha_k\,E_{(k)}{}^M{}_N + \Lambda\,\delta^M_N = \kappa\,T^M{}_N, \qquad E_{(k)}{}^M{}_N = -\frac{1}{2^{k+1}}\,\delta^{Ma_1b_1\cdots a_kb_k}_{Nc_1d_1\cdots c_kd_k}\,R^{c_1d_1}{}_{a_1b_1}\cdots R^{c_kd_k}{}_{a_kb_k},$$

$E_{(1)}$ being the Einstein tensor. The author's weights are

$$\alpha_k = w_k H^{2-2k}, \qquad w = (w_1,w_2,w_3) = \Big(1, \frac{1}{32}, \frac{1}{1152}\Big), \qquad \Lambda = -10H^2$$

(the author's notebook writes $\Lambda = 10$ with the opposite sign convention for Λ). UM-2 keeps these values.

For a pair-diagonal curvature (DERIVED). When $R^{ab}{}_{cd}$ is pair-diagonal (the order-0 metric),

$$\mathcal{L}_{(k)} = 2^k k! \sum_{\text{k disjoint pairs}}\ \prod_{\text{pairs}} K_{ab}, \qquad E_{(k)}{}^h{}_h = -2^{k-1}k! \sum_{\text{k disjoint pairs avoiding } h}\ \prod_{\text{pairs}} K_{ab},$$

which was checked against the Einstein tensor from the Ricci tensor ($k = 1$), against the explicit Gauss-Bonnet tensor ($k = 2$), against the Bianchi identity $\nabla_M E_{(k)}{}^M{}_N = 0$ ($k = 1,2,3$), against the Euler-Lagrange equations of the action with a lapse ($k = 1,2,3$) and against the record's equations (all exact).

5.2 The block stabiliser (ASSUMED)

A source term

$$S_{stab} = -\int d^8x\,\sqrt{|g|}\,\rho_b(f), \qquad \kappa\,\rho_b(f) = \kappa\rho^*_b\Big(\frac{f_*}{f}\Big)^8,$$

whose energy-momentum tensor is that of a vacuum energy of the block,

$$T^M{}_N[\text{stab}] = \mathrm{diag}\big(p_0,p_3,p_3,p_3,-\rho_b,p_t,p_t,p_t\big), \qquad p_3 = -\rho_b, \qquad p_0 = p_t = +\rho_b.$$

It is conserved ($\dot\rho_b = -3A(\rho_b + p_3) - 4h(\rho_b + p_0) = -8h\rho_b$ gives $\rho_b \propto f^{-8}$), it is four-dimensionally Lorentz invariant ($p_3 = -\rho_b$) and isotropic in the block ($p_0 = p_t$), as the field equations require (section 8.1), and it does not depend on x_0. The f^{-8} law is that of an eight-dimensional energy density whose only scale is the size of the block (a Casimir-type or flux energy). Its normalisation is fixed by the static block (section 8.2): $\kappa\rho^*_b = (26 + 4\sqrt{51})H^2$ at $f_* = k_*^{-1/2}$. Its microscopic origin is OPEN (a Casimir energy of the fields of the cell or a flux are natural candidates; neither was computed in UM-2).

5.3 The dirac16complex field

5.3.1 The Clifford algebra of the author (AUTHOR, DERIVED verification)

The author's real 16×16 matrices are built as follows ($h=1,2,3$; $p,q=1,\dots,4$; ϵ the Levi-Civita symbol; I_n the identity):

$$Q_a(h,p,q)=\epsilon_{hpq4},\quad Q_b(h,p,q)=\delta_{p4}\delta_{qh}-\delta_{ph}\delta_{q4},\quad s_h=Q_a-Q_b,\quad t_h=Q_a+Q_b,$$

$$\zeta_0=I_8,\quad \zeta_{7-h}=\begin{pmatrix}0 & t_h\\ -t_h & 0\end{pmatrix},\quad \zeta_h=\begin{pmatrix}0 & s_h\\ s_h & 0\end{pmatrix},\quad \zeta_7=\zeta_1\zeta_2\zeta_3\zeta_4\zeta_5\zeta_6,\quad \sigma=\begin{pmatrix}0 & I_4\\ I_4 & 0\end{pmatrix},$$

$$\tau_0=I_8,\quad \tau_A=\zeta_A\ (A=1..7),\quad \bar\tau_A=\sigma\,\tau_A^T\sigma,\qquad \gamma^{x_A}\equiv T16[A]=\begin{pmatrix}0 & \bar\tau_A\\ \tau_A & 0\end{pmatrix}\quad (A=0,\dots,7).$$

They satisfy (DERIVED, exact; 31/31 checks)

$$\{\gamma^a,\gamma^b\}=2\eta^{ab}I_{16},\qquad \tfrac12(\tau_A\bar\tau_B+\tau_B\bar\tau_A)=\eta_{AB}I_8,\qquad \sigma=\tau_1\tau_2\tau_3=\tau_4\tau_5\tau_6\tau_7.$$

The charge-conjugation matrix, the chirality, the Krein metric and the generators are

$$C=\gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}=\begin{pmatrix}-\sigma & 0\\ 0 & \sigma\end{pmatrix},\quad \Gamma=\gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7}=\mathrm{diag}(-I_8,I_8),\quad B=-iC\gamma^{x_4},\quad S^{ab}=\tfrac14[\gamma^a,\gamma^b],$$

with $C^T=C$, $C^2=1$, $(C\gamma^a)^T=-C\gamma^a$, $B^\dagger=B$, $B^2=1$ and B of signature $(8,8)$. The 16-dimensional representation is irreducible under Pin(4,4) (commutant of dimension 1), the double cover of $O(4,4)$, and it splits under Spin(4,4), the double cover of $SO(4,4)$, into two inequivalent irreducible 8-dimensional representations, the eigenspaces $\Gamma=\mp1$ (commutant of dimension 2, intertwiner of dimension 0).

5.3.2 The Lagrangian and the canonical spin connection

Ψ has 16 complex anticommuting (Grassmann) components and $\bar\Psi=\Psi^\dagger C$. The Lagrangian density is

$$\mathcal{L}_\Psi=\sqrt{|g|}\Big[\tfrac12\big(\bar\Psi\gamma^M D_M\Psi-(D_M\bar\Psi)\gamma^M\Psi\big)-m\,\bar\Psi\Psi-\tfrac{\lambda}{2}(\bar\Psi\Psi)^2\Big],$$

$$\gamma^M=e_a{}^M\gamma^a,\quad D_M\Psi=\partial_M\Psi+\Omega_M\Psi,\quad D_M\bar\Psi=\partial_M\bar\Psi-\bar\Psi\Omega_M,\quad \Omega_M=\tfrac12\,\omega_{Mab}S^{ab},$$

$$\omega_M{}^a{}_b=e^a{}_N\big(\partial_M e_b{}^N+\Gamma^N{}_{ML}\,e_b{}^L\big),\qquad \omega_{Mab}=\eta_{ac}\,\omega_M{}^c{}_b,$$

the canonical (metric-compatible, torsion-free) spin connection: it makes γ^N covariantly constant, $\partial_M\gamma^N+\Gamma^N{}_{ML}\gamma^L+[\Omega_M,\gamma^N]=0$ (DERIVED). For the order-0 metric (DERIVED, exact, for every a_4, b_4 and on both halves)

$$\gamma^M\Omega_M=\frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0}+\Big(\frac{\dot b_4}{b_4}-\frac{\dot a_4}{2}\Big)\gamma^{x_4},$$

which vanishes only in flat 4+4 space ($H=0$ with constant a_4, b_4): the coupling of Ψ to gravity is non-trivial. The field equation is

$$\gamma^M D_M\Psi=\big(m+\lambda\,\bar\Psi\Psi\big)\Psi.$$

The mass parameter is the author's $m=3HM$; UM-2 uses $M=1/3$, i.e. $m=H$ (ASSUMED, the value of the Kohn-Sham record). λ (dimension mass^{-6}) is the contact self-coupling. It lets the dark matter annihilate into the fold bands, so the cosmology bounds it: $G_\lambda=\lambda H\tilde I_\lambda/(2f^4v_t)\le 2.3\times10^{-15}$ GeV^{-2} (ASSUMED, section 11.2), i.e. $\lambda\le1.4\times10^{-10}H^{-6}$ for $v_t=H^{-3}$; the value $\lambda_B=7\times10^{-5}H^{-6}$ of the Kohn-Sham tests (sections 9 and 9.1) is a numerical test value.

5.3.3 The two halves Ψ_A and Ψ_B and the fold parity (DERIVED)

B commutes with every four-dimensional Lorentz generator $\gamma^a\gamma^b$ ($a,b\in\{x_1,..,x_4\}$) and with C, so

$$\Psi=\Psi_A+\Psi_B,\qquad B\Psi_A=-\Psi_A,\qquad B\Psi_B=+\Psi_B$$

splits Ψ into two four-dimensionally Lorentz-invariant halves of 8 components, and $\bar\Psi\Psi=\bar\Psi_A\Psi_A+\bar\Psi_B\Psi_B$ does not mix them. In the good sector (no extra-time momenta) every interaction of UM-2 commutes with B (DERIVED with the author's matrices: B commutes with C, γ^a and $C\gamma^a$ for $a=x_0,\dots,x_4$ and anticommutes with them for $a=x_5,x_6,x_7$), so the numbers N_A and N_B are separately conserved; extra-time momenta would mix the halves. Under the fold reflection R the modes are even or odd; the bulk Lagrangian is R-symmetric.

5.4 The Standard-Model gauge fields (ASSUMED structure, STANDARD content)

The gauge group is $SU(3)_c \times SU(2)_L \times U(1)_Y$ with eight-dimensional gauge fields G^α_M ($\alpha = 1..8$), W^a_M ($a = 1,2,3$) and B_M:

$$S_{gauge} = -\frac{1}{4}\int d^8x\sqrt{|g|}\, g^{MP}g^{NQ}\Big(\frac{1}{\hat g_s^2}G^\alpha_{MN}G^\alpha_{PQ} + \frac{1}{\hat g^2}W^a_{MN}W^a_{PQ} + \frac{1}{\hat g'^2}B_{MN}B_{PQ}\Big),$$

$$G^\alpha_{MN} = \partial_M G^\alpha_N - \partial_N G^\alpha_M + f^{\alpha\beta\gamma}G^\beta_M G^\gamma_N, \quad W^a_{MN} = \partial_M W^a_N - \partial_N W^a_M + \epsilon^{abc}W^b_M W^c_N, \quad B_{MN} = \partial_M B_N - \partial_N B_M.$$

The four-dimensional couplings are $g_s^2 = \hat g_s^2/V_\perp$, $g^2 = \hat g^2/V_\perp$, $g'^2 = \hat g'^2/V_\perp$, where $V_\perp$ is the proper volume of the block per unit 4-volume (section 7). In this geometric normalisation the couplings stand in front of the kinetic terms and not in the covariant derivatives; the dimensions are $[G_M] = [W_M] = [B_M] = 1$, $[\hat g^2] = -4$ (mass units), and the four-dimensional couplings are dimensionless. From here on $W^\pm$, Z, A and every formula that shows $\hat g$, $\hat g'$ explicitly refer to the canonically normalised fields $W^a/\hat g$, $B/\hat g'$, $G^\alpha/\hat g_s$ (written with the same letters). The physical fields are

$$W^\pm_M = \frac{W^1_M \mp iW^2_M}{\sqrt{2}}, \qquad Z_M = \cos\theta_W\, W^3_M - \sin\theta_W\, B_M, \qquad A_M = \sin\theta_W\, W^3_M + \cos\theta_W\, B_M, \qquad \tan\theta_W = \frac{g'}{g}.$$

W^+ carries the electric charge $+e$ and W^- the charge $-e$ ($[T_3, T_\pm] = \pm T_\pm$, $Y(W) = 0$; W^- is the antiparticle of W^+); Z and the photon A are neutral (DERIVED). A fermion of weak isospin T_3 and hypercharge Y couples to A with $e(T_3 + Y) = eQ$, $e = gg'/\sqrt{g^2+g'^2}$, and to Z with $\sqrt{g^2+g'^2}\,(T_3 - Q\sin^2\theta_W)$ (DERIVED). The non-abelian terms of $W^a_{MN}W^a_{PQ}$ contain the vertices $W^+W^-\gamma$ (strength e), W^+W^-Z (strength $g\cos\theta_W$) and the quartic $W^+W^-W^+W^-$, $W^+W^-\gamma\gamma$, $W^+W^-Z\gamma$, W^+W^-ZZ; every vertex conserves electric charge, so a single W^+ or W^- always comes with a charged partner.

5.5 The Standard-Model fermions (three generations; ASSUMED structure, INVENTED boundary assignment)

Each multiplet is an eight-dimensional spinor of the same Clifford algebra (16 components times its gauge indices). There are three generations, $i = 1,2,3$, with the same quantum numbers:

field ($i = 1,2,3$)	members	$SU(3)_c$	$SU(2)_L$	Y	electric charges $Q = T_3 + Y$	4D zero mode
Q_i	$(u,d)_L$, $(c,s)_L$, $(t,b)_L$	3	2	1/6	up-type 2/3, down-type $-1/3$	$\Gamma_4 = +1$
u_{Ri}	u_R, c_R, t_R	3	1	2/3	2/3	$\Gamma_4 = -1$
d_{Ri}	d_R, s_R, b_R	3	1	$-1/3$	$-1/3$	$\Gamma_4 = -1$
L_i	$(\nu_e, e)_L$, $(\nu_\mu, \mu)_L$, $(\nu_\tau, \tau)_L$	1	2	$-1/2$	neutrino 0, charged lepton -1	$\Gamma_4 = +1$
e_{Ri}	e_R, μ_R, τ_R	1	1	-1	-1	$\Gamma_4 = -1$
ν_{Ri}	three right-handed neutrinos	1	1	0	0	$\Gamma_4 = -1$

with the action

$$S_f = \sum_{i=1}^{3}\sum_{\psi}\int d^8x\sqrt{|g|}\Big[\tfrac{1}{2}\big(\bar\psi_i\gamma^M D_M\psi_i - (D_M\bar\psi_i)\gamma^M\psi_i\big) - \mu_{\psi i}\bar\psi_i\psi_i\Big],$$

$$D_M = \partial_M + \Omega_M - iG^\alpha_M T^\alpha - iW^a_M\frac{\tau^a}{2} - iYB_M,$$

$\psi \in \{Q, u_R, d_R, L, e_R, \nu_R\}$, the same canonical spin connection Ω_M as for Ψ and a constant bulk mass $\mu_{\psi i}$ per multiplet. The bulk masses are diagonal in the generation index, and the doublet bulk mass is the same for the three generations of a sector (ASSUMED):

$$\mu_{Li} = \mu_L, \qquad \mu_{Qi} = \mu_Q \qquad (i = 1,2,3),$$

while the singlet bulk masses $\mu_{e_R i}$, $\mu_{\nu_R i}$, $\mu_{u_R i}$, $\mu_{d_R i}$ differ from generation to generation. The 4D chirality is $\Gamma_4 = i\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$.

Boundary conditions. In 16-component form, with the selection projector

$$P_{sel} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-i\gamma^{x_5}\gamma^{x_6}}{2},$$

which commutes with every four-dimensional Lorentz generator (DERIVED), the conditions at the fold ($y=0$) and at the tip ($y=-L$) are

- doublets (Q, L; $\mu_\psi > 0$): fold $P_{sel}(1-\gamma^{x_0})\psi = 0$ and $(1-P_{sel})(1+\gamma^{x_0})\psi = 0$; tip $(1-\gamma^{x_0})\psi = 0$;
- singlets (u_R, d_R, e_R, ν_R; $\mu_\psi < 0$): fold $P_{sel}(1+\gamma^{x_0})\psi = 0$ and $(1-P_{sel})(1-\gamma^{x_0})\psi = 0$; tip $(1+\gamma^{x_0})\psi = 0$.

These are self-adjoint boundary conditions of the Dirac operator (DERIVED). The result (DERIVED, section 6): every doublet field has exactly one four-dimensional Weyl zero mode with $\Gamma_4 = +1$ and every singlet field exactly one with $\Gamma_4 = -1$, all localised at the fold and all in the two blocks selected by P_{sel}, which lie in the $B = -1$ half. The zero-mode spectrum is three Standard-Model generations, each with its ν_R. Writing every zero mode as a left-handed Weyl field (a right-handed field of hypercharge Y as a left-handed one of hypercharge $-Y$), with n_c colours and n_w weak components, the anomaly coefficients of one generation are (DERIVED, exact fractions)

$$\mathcal{A}_{SU(3)^2U(1)} = \sum_{\text{triplets}} n_w Y = 0, \qquad \mathcal{A}_{SU(2)^2U(1)} = \sum_{\text{doublets}} n_c Y = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

$$\mathcal{A}_{U(1)^3} = \sum n_c n_w Y^3 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = 0, \qquad \mathcal{A}_{grav^2U(1)} = \sum n_c n_w Y = 1 - 2 + 1 - 1 + 1 = 0,$$

and $\mathcal{A}_{SU(3)^3}$ (left-handed minus right-handed triplets) $= 2 - 1 - 1 = 0$; the twelve $SU(2)$ doublets of three generations are an even number (no Witten anomaly). The anomalies cancel generation by generation; three lepton generations with a single quark generation would leave $\mathcal{A}_{SU(2)^2U(1)} = -1$ and $\mathcal{A}_{U(1)^3} = 3/2$, so the second and third quark generations are required together with the second and third lepton generations (DERIVED).

QCD (STANDARD). The gluons are the massless x_0 zero modes of G^α_M. Below the QCD scale the quarks are confined into hadrons; the top quark decays (into W^+b) before it can form one. The baryons of the cosmology are the protons (uud, charge $+1$) and neutrons (udd, charge 0), whose masses $m_p = 938.3$ MeV and $m_n = 939.6$ MeV come from QCD; they are inputs of the cosmology. The QCD scale depends on the block through $g_s^2 = \hat{g}_s^2/V_\perp$.

5.6 The tip: electroweak symmetry breaking without a Higgs particle (INVENTED for this geometry)

At the regular tip a non-linear sigma field $\Sigma(x) \in SU(2)$, $\Sigma^\dagger\Sigma = 1$, transforming as $\Sigma \to U_L\,\Sigma\,U_Y^\dagger$ with $U_Y = \exp(i\theta\tau^3/2)$, carries the electroweak symmetry breaking:

$$S_{tip} = \int_{y=-L} d^7x\,\sqrt{|g_{tip}|}\Big[-\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \sum_{i,j=1}^{3}\big(\bar{L}_i\,\Sigma\,(\Lambda_\ell)_{ij}\,\ell_{Rj} + \bar{Q}_i\,\Sigma\,(\Lambda_q)_{ij}\,q_{Rj} + \text{h.c.}\big)\Big],$$

$$D_\mu\Sigma = \partial_\mu\Sigma - iW_\mu^a\frac{\tau^a}{2}\Sigma + iB_\mu\,\Sigma\frac{\tau^3}{2}, \qquad \ell_{Rj} = (\nu_{Rj}, e_{Rj}), \quad q_{Rj} = (u_{Rj}, d_{Rj}),$$

$$(\Lambda_\ell)_{ij} = \mathrm{diag}\big((\lambda_\nu)_{ij}, (\lambda_e)_{ij}\big), \qquad (\Lambda_q)_{ij} = \mathrm{diag}\big((\lambda_u)_{ij}, (\lambda_d)_{ij}\big),$$

with four complex 3×3 tip coupling matrices λ_ν, λ_e, λ_u, λ_d (the diagonal acts on the isospin components: the upper component of $\Sigma^\dagger L_i$ couples to ν_R, the lower one to e_R). Σ has no radial mode, so there is no Higgs particle; its three would-be Goldstone modes become the longitudinal W^+, W^- and Z. Every term is $SU(2)_L \times U(1)_Y$ invariant, the Yukawa terms for arbitrary complex generation matrices (DERIVED with random group elements and random matrices, deviation 1.1×10^{-14}). In the unitary gauge $\Sigma = 1$ the first term is

$$-\frac{v_\Sigma^2}{8}\Big[\hat{g}^2\big(W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu}\big) + (\hat{g}^2 + \hat{g}'^2)Z_\mu Z^\mu\Big] = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}^2\,W_\mu^+ W^{-\mu} + (\hat{g}^2 + \hat{g}'^2)Z_\mu Z^\mu\Big],$$

which gives W^+ and W^- one common mass (they are particle and antiparticle), the Z a mass larger by $1/\cos\theta_W$ at leading order (custodial symmetry), and leaves the photon massless. The second term mixes the doublet and singlet zero modes at the tip and gives the fermions their masses and their mixing (section 6.2).

5.7 The portal interactions of dirac16complex (INVENTED, gauge invariant)

Ψ is a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$, so it couples to the Standard Model only through gauge-invariant composite operators. UM-2 contains three of them (the names I11 and I12 are those of version 1; I13 is new in version 3).

I11: the vector contact portal.

$$S_{I11} = \int d^8x\,\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}\,(i\bar\Psi\gamma^M\Psi)\sum_{i=1}^{3}\sum_{\psi} q_\psi\,(i\bar\psi_i\gamma_M\psi_i), \qquad q_Q = q_{u_R} = q_{d_R} = \tfrac{1}{3}, \quad q_L = q_{e_R} = q_{\nu_R} = -1,$$

summed over the multiplets of the three generations with the $B-L$ charges q_ψ (each term is separately gauge invariant and generation blind; with the right-handed neutrinos $B-L$ is free of gauge anomalies); $\Lambda_\star$ is the common scale of the bulk portals, and c_V is dimensionless ($[\bar\Psi\gamma^M\Psi] = [\bar\psi\gamma_M\psi] = 7$). It couples both halves of Ψ, including the fold bands, to the fermion currents. On the zero modes it becomes the four-dimensional contact interaction between a state o of Ψ (an orbital (a_o, b_o) normalised on one half of the cell, $\int_{-L}^{0}(a_o^2+b_o^2)\,dy = 1$) and a chirality component f of a Standard-Model fermion with the normalised zero-mode profile $\varrho_f(y)$ ($\int_{-L}^{0}\varrho_f\,dy = 1$; $\varrho_c(y) = (2c/L)e^{2cy/L}/(1-e^{-2c})$ for the localisation c),

$$\mathcal{L}_4 = G_V\,q_f\,\tilde I_{of}\,(i\bar\chi_o\gamma^\mu\chi_o)(i\bar f\gamma_\mu f), \qquad \tilde I_{of} = \int_{-L}^{0}(a_o^2+b_o^2)\,\varrho_f(y)\,e^{-3Hy}\,dy, \qquad G_V = \frac{c_V H}{2f^4 v_t \Lambda_\star^6}$$

($\tilde I$ in units of H, y in units of $1/H$; v_t the coordinate volume of the extra times; DERIVED: $\tilde I_{of}$ is exactly the response $d\varepsilon_o/de$ of the Kohn-Sham level to the mean field $e\,\varrho_f(y)e^{-3Hy}$). Its strength must be small to keep the massless fold bands out of equilibrium; version 3 fixes the benchmark (ASSUMED) by their freeze-in: the Standard-Model plasma produces fold-band pairs, $f\bar f \to \Psi_{fold}\bar\Psi_{fold}$, with the spin-averaged cross-section $G_f^2 s/(12\pi)$ per Dirac fermion f and Dirac fold band ($G_f = G_V q_f \tilde I_f$; half of it for one chirality), i.e. at the rate (pairs per volume and time; Maxwell-Boltzmann statistics, massless particles; N_c the colour multiplicity of the Weyl component f)

$$\gamma = \frac{4}{\pi^5}\,N_\chi\,G_V^2\,T^8\sum_f N_c\,(q_f\tilde I_f)^2, \qquad N_\chi = 2 \text{ (the fold bands of } \Psi_A \text{ and } \Psi_B),$$

each pair carrying the mean energy $8T$; produced from the highest temperature of the hot Universe, $T_{max} = 10$ TeV (ASSUMED), they add $\Delta N_{eff} = 0.001$ for $G_V = 4.38\times10^{-17}$ GeV^{-2} ($3.8\times10^{-12}\,G_F$), without ever reaching equilibrium (DERIVED, section 11.2).

I12: the gauge-kinetic scalar portal.

$$S_{I12} = -\int d^8x\,\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_G\,G^\alpha_{MN}G^{\alpha MN} + c_W\,W^a_{MN}W^{aMN} + c_B\,B_{MN}B^{MN}\Big]$$

(field strengths in the geometric normalisation of section 5.4; c_G, c_W, c_B dimensionless, since $[\bar\Psi\Psi] = 7$ and $[G_{MN}G^{MN}] = 4$). With $W^aW^a = 2W^+_{MN}W^{-MN} + W^3W^3 + \ldots$ and $W^3 = \cos\theta_W Z + \sin\theta_W A$, $B = -\sin\theta_W Z + \cos\theta_W A$, it couples $\bar\Psi\Psi$ to the pairs W^+W^-, ZZ, $Z\gamma$, $\gamma\gamma$ and gluon pairs. The scalar density of every chiral zero mode vanishes identically (its lower component is zero), so **I12 does not touch the massless fold bands**; it couples the massive states of Ψ, in particular the dark matter, which it lets annihilate into gauge-boson pairs (section 11). In the QCD vacuum the gluon condensate $\langle G^\alpha G^\alpha\rangle$ makes I12 a uniform mass shift of Ψ. For the dark matter it becomes $G_i\bar\chi\chi F^i_{\mu\nu}F^{i\mu\nu}$ (canonical fields) with $G_i = \mathcal{O}\,c_i g_i^2/\Lambda_\star^3$, $\mathcal{O} = \partial\varepsilon_{DM}/\partial m = 0.7072$ (DERIVED). Its benchmark strength (ASSUMED, version 3) is set by unitarity: $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = \pi/(2M_{DM}^3) = (38.2\text{ TeV})^{-3}$ keeps the effective annihilation cross-section below the ${}^3P_0$ partial-wave bound up to $\sqrt{s} = 2.40\,M_{DM}$ (section 11); with universal coefficients $c_G = c_W = c_B = c$ and the gauge couplings at the annihilation scale $2M_{DM}$ (one loop: $g_s^2 = 0.779$, $g^2 = 0.378$, $g'^2 = 0.138$) this is $c/\Lambda_\star^3 = 1.106\times10^{-14}$ GeV^{-3}, i.e. $\Lambda_\star/c^{1/3} = 44.9$ TeV, and G_g, G_W, $G_B = 6.09$, 2.96, 1.08×10^{-15} GeV^{-3}.

I13: the x_0-chiral vector portal (version 3).

$$S_{I13} = \int d^8x\,\sqrt{|g|}\,\frac{c_A}{\Lambda_\star^6}\sum_{M\neq x_0}(i\bar\Psi\gamma^M P_b\Psi)\,J^B_M, \qquad P_b = \frac{1-\gamma^{x_0}}{2}, \qquad J^B_M = \frac{1}{3}\sum_{i=1}^{3}\sum_{\psi\in\{Q_i,\,u_{Ri},\,d_{Ri}\}} i\bar\psi\gamma_M\psi$$

(γ^{x_0} the frame matrix; c_A dimensionless; the sum runs over the seven directions $M = x_1, \dots, x_7$). Properties (DERIVED):

- for $M \neq x_0$ the bilinear $i\bar\Psi\gamma^M P_b\Psi$ is Hermitian: γ^{x_0} anticommutes with every γ^M, $M \neq x_0$, so $P_a\gamma^M = \gamma^M P_b$ with $P_a = (1+\gamma^{x_0})/2$, and C anticommutes with γ^{x_0}, so $CP_bC^{-1} = P_a$, which makes $(iC\gamma^M P_b)^\dagger = iC\gamma^M P_b$ for the real matrices of the author. For $M = x_0$ it is not: γ^{x_0} commutes with P_b, and $(iC\gamma^{x_0}P_b)^\dagger = iC\gamma^{x_0}P_a$; the component $M = x_0$ is therefore left out (it would make the action complex; $J^B_{x_0}$ vanishes on the quark zero modes anyway). Its four-dimensional components $\mu = x_1, \dots, x_4$ form a Lorentz vector (γ^{x_0} commutes with the four-dimensional Lorentz generators), and $C\gamma^\mu P_b$ commutes with B for $\mu = x_1, \dots, x_4$, so I13 does not mix the halves Ψ_A, Ψ_B in four dimensions; for the extra times ($M = x_5, x_6, x_7$) $C\gamma^M P_b$ anticommutes with B, which is harmless because $J^B_{x_5}$, $J^B_{x_6}$, $J^B_{x_7}$ vanish on the zero modes (which lie in the half $B = -1$);
- in the block form of the Kohn-Sham reduction $\gamma^{x_0} = \sigma_3$, so P_b projects on the lower component b of an orbital; the fold bands are $(a, 0) = (e^{f\int M\,dy}, 0)$, so **I13 vanishes identically on the fold bands** and cannot produce dark radiation; it couples the massive states, in particular the dark matter (the lowest odd level, $a(0) = 0$, $b \neq 0$);
- J^B is the baryon-number current, a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$ (term by term), conserved in perturbation theory.

On the zero modes it becomes

$$\mathcal{L}_4 = G_A\, q_f\, \tilde I^{(b)}_{of}\, (i\bar\chi_o\gamma^\mu\chi_o)(i\bar f\gamma_\mu f), \qquad \tilde I^{(b)}_{of} = \int_{-L}^{0} b_o^2\, \varrho_f(y)\, e^{-3Hy}\, dy, \qquad G_A = \frac{c_A H}{2f^4 v_t \Lambda_\star^6}, \qquad q_f = \frac{1}{3}.$$

The dark matter annihilates through it in the s-wave ($^3S_1$) into quark pairs: for one Weyl component f with $G_f = G_A q_f \tilde I^{(b)}_f$ and N_c colours (spin averaged, the quark masses neglected)

$$\sigma(\chi\bar\chi \to f\bar f) = \frac{N_c G_f^2\, s\,(1 + 2M^2/s)}{24\pi\beta}, \qquad \beta = \sqrt{1 - 4M^2/s}, \qquad \sigma v \to \frac{N_c G_f^2 M^2}{2\pi} \quad (v \to 0),$$

which partial-wave unitarity bounds by $\sigma \le 3\pi/(4p^2)$ (the three $J = 1$ states of $^3S_1$ averaged over the four spin states). With the overlaps of the dark-matter orbital, $\sum_f N_c (q_f \tilde I^{(b)}_f)^2 = 16.35$, the two dark-matter species (with the I12 benchmark) freeze out with $\Omega h^2 = 0.060$ each for $G_A = 1.278 \times 10^{-9}$ GeV^{-2} ($1.1 \times 10^{-4}\, G_F$), the cross-section capped at the unitarity limit, which it reaches only above $v = 0.51$ (22 % of the freeze-out rate) (DERIVED, section 11.2).

Why the baryon-number current (DERIVED; the elastic rate is an ESTIMATE). With the $B - L$ current instead of J^B, the annihilation $\chi\bar\chi \leftrightarrow \nu_R\bar\nu_R$ would bring the three right-handed neutrinos into equilibrium above the freeze-out (from zero abundance they reach 0.9997 of the equilibrium number by $M/T = 40$; the number-changing rate falls below the Hubble rate at $M/T = 16.1$, elastic scattering on the dark matter, $\Gamma_{el} = \sum_{A,B}(n_\chi + n_{\bar\chi}) G^2 E^2/(2\pi)$ with $E \simeq 3T$ for a Weyl ν_R on a heavy Dirac fermion, at $M/T = 24.4$, $T = 1.82$ TeV), and they would decouple as three Weyl species adding $\Delta N_{eff} = 3(10.75/g_s)^{4/3} F_h = 0.144$ ($g_s = 105.7$), i.e. $N_{eff} = 3.19$, 2.5 standard deviations above the ACT DR6 value 2.86 ± 0.13. With J^B the right-handed neutrinos keep only their tip coupling and their I11 coupling ($q_{\nu_R} = -1$): the annihilation into ν_R pairs then goes through I11 alone, at 2.0×10^{-17} of the I13 rate into quarks, and they are never thermalised.

Removed in version 2: the chiral tip Z portal. Version 1 contained a third portal,

$$S_{I9} = \int_{y=-L} d^7x\, \sqrt{|g_{tip}|}\, \big(c_Z + c'_Z\, \varsigma\big)\, (i\bar\Psi\gamma^\mu P_+\Psi)\, \frac{i v_\Sigma^2}{2} \mathrm{tr}\big(\Sigma^\dagger D_\mu \Sigma\, \tau^3\big), \qquad P_+ = \frac{1 + \Gamma_4}{2},$$

($\varsigma = \pm 1$ distinguishing the two sides of a tip; in the unitary gauge $\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu \Sigma \tau^3) = \frac{v_\Sigma^2}{2}\sqrt{\hat g^2 + \hat g'^2}\, Z_\mu$), whose even part c_Z made the Ψ_A fold band (two Weyl species) couple to the Z like two neutrinos and so stand in for the missing ν_μ, ν_τ of one generation. Its algebra is DERIVED and remains true: P_+ is a Hermitian projector commuting with the 4D Lorentz generators and with $C\gamma^\mu$; $\Gamma_4 = -B\gamma^{x_0}$, so $P_+ = (1 - B\gamma^{x_0})/2$, which is $(1+\gamma^{x_0})/2$ on Ψ_A and $(1-\gamma^{x_0})/2$ on Ψ_B and annihilates Ψ_B at the tip (every orbital satisfies $(1-\gamma^{x_0})\Psi = 0$ there): **no P_+ tip current and no tip operator built on $\bar\Psi\Psi$ reaches Ψ_B** (a non-chiral or P_- tip current would). With three neutrinos the portal has no role left, and both of its parts are bounded or disfavoured by the data (DERIVED, section 10.7):

- its even part gives the Z an invisible width $N_\nu = 3 + 2r^2$ (r the Z coupling of the fold band relative to a neutrino's), bounded by the measured $N_\nu = 2.9963 \pm 0.0074$ to $r^2 < 0.0056$; even at that bound the fold band would stay in equilibrium down to 14.7 MeV and add

$$\Delta N_{eff} = n_W \Big(\frac{10.75}{g_s}\Big)^{4/3} F_h, \qquad F_h = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/(1+h_\nu)\Big]^4 = 1.0095$$

  ($n_W = 2$ Weyl species decoupling when the plasma has g_s entropy degrees of freedom, here $g_s = 10.75$; F_h corrects for the heating of the neutrinos by e^+e^- annihilation, which an earlier-decoupled species does not receive: $T_\nu/T_\gamma = 0.71664$ and the comoving heating $h_\nu = 0.00165$ of the common-temperature decoupling of section 10.4), i.e. $\Delta N_{eff} = 2.02$, far above the measured $N_{eff} = 2.99 \pm 0.17$ (Planck 2018) and 2.86 ± 0.13 (ACT DR6) (STANDARD);
- its oriented part c'_Z (odd under the fold reflection) would let the heavy odd Ψ_A states decay, $\Psi_A^{odd} \to \Psi_A^{fold} + Z$, and by the inverse decays and the exchange process $\Psi_A^{fold} Z \to \Psi_A^{fold} Z$ keep the fold band in equilibrium until the odd states or the Z bosons disappear (at $g_s = 105.75$ or 86.25), adding $\Delta N_{eff} = 0.096$ to 0.126 (ESTIMATE), i.e. $N_{eff} = 3.14$ to 3.17, 2.1 to 2.4 standard deviations above the ACT DR6 value (disfavoured).

UM-2 therefore sets $c_Z = c'_Z = 0$ (ASSUMED). Consequences: $N_\nu = 3$ exactly; neither fold band is ever thermalised (no tip operator, zero scalar density for I12, the weak benchmark of I11, which produces them only by freeze-in with $\Delta N_{eff} = 0.001$, and the bound on the self-interaction of section 11.2); both halves keep an exact fold parity, so the lightest odd states of Ψ_A and of Ψ_B are both stable (section 11).

Not included, and why. A dipole portal $\bar\Psi S^{MN} \Psi B_{MN}$ would couple the fold bands to the photon and thermalise them (too much dark radiation). No gauge-invariant operator couples the neutral Ψ to a single W^+ or W^-; $W^\pm$ enter only in pairs (I12) or with a charged fermion pair (the covariant derivatives of I11).

5.8 The catalogue of interactions

#	interaction	fields	equation (vertex)	role in UM-2
1	Einstein-Lovelock self-interaction	g	$\sum_k \alpha_k \mathcal{L}_{(k)} - 2\Lambda$	background, block dynamics, 4D Newton constant $\propto 1/G(k)$
2	gravity of every field	g with all	$\sqrt{\lvert g\rvert}$, g^{MN}, Ω_M in each action	universal coupling; Ψ through the canonical spin connection
3	block stabiliser	g	$-\sqrt{\lvert g\rvert}\rho_b(f)$	static, stable block with attractive 4D gravity
4	Ψ self-interaction	Ψ	$-\frac{\lambda}{2}(\bar\Psi\Psi)^2$	Kohn-Sham mean fields M_{eff}, v
5	strong interaction	G, the six quarks	$-G_M^\alpha\,(i\bar q\gamma^M T^\alpha q)$, $f^{\alpha\beta\gamma}GGG$, $GGGG$	confinement, p and n
6	electromagnetic interaction	A, charged fermions, $W^\pm$	$-eQ\,A_M\,(i\bar\psi\gamma^M\psi)$; $W^+W^-\gamma$ (strength e)	CMB photons, recombination, Thomson drag
7	weak charged current	$W^\pm$, the three lepton and quark generations	$-\frac{g}{\sqrt2}W^+_M J^M_{CC}$ and h.c.; $J^M_{CC} = \sum_\alpha i\bar\nu_\alpha\gamma^M P_+ \ell_\alpha + \sum_{ab} V_{ab}\, i\bar u_a\gamma^M P_+ d_b$; $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$	β decay, $n \leftrightarrow p$, neutrinos produced and detected in flavour states, flavour conversion
8	weak neutral current	Z, all fermions	$-\sqrt{g^2+g'^2}\,Z_M \times(T_3 - Q\sin^2\theta_W)\,i\bar\psi\gamma^M\psi$, flavour diagonal in every basis	neutrino decoupling, Z widths

#	interaction	fields	equation (vertex)	role in UM-2
9	gauge self-interactions	$W^{\pm}$, Z, γ	W^+W^-Z, $W^+W^-W^+W^-$, $W^+W^-\gamma\gamma$, $W^+W^-Z\gamma$, W^+W^-ZZ	electroweak consistency
10	$W^{\pm}$, Z masses	Σ, W^a ($W^{\pm}$, W^3), B	$-\frac{v_\Sigma^2}{4}\mathrm{tr}(D\Sigma^\dagger D\Sigma)$ at the tip	x_0 profiles bent at the tip
11	fermion masses and mixing	Σ, L_i, ℓ_{Rj}, Q_i, q_{Rj}	$\sum_{ij}\left(\bar{L}_i\Sigma(\Lambda_\ell)_{ij}\ell_{Rj}+\bar{Q}_i\Sigma(\Lambda_q)_{ij}q_{Rj}\right)$ at the tip	tip mixing of the chiral zero modes: masses, PMNS, CKM
12	neutrino forward scattering (7 and 8 at low energy)	ν with e, n, ν	$V_{CC}=\sqrt{2}G_F n_e$ on ν_e; V_{NC} flavour universal; $\nu\nu$ term	MSW conversion (Sun, Earth, supernovae), early-Universe potentials, fast flavour instability
13	Ψ - fermion currents	Ψ, all fermions	I11 ($B-L$)	freeze-in of the fold bands ($\Delta N_{eff}=0.001$); Kohn-Sham mean fields
14	Ψ - gauge pairs	Ψ, W^+W^-, ZZ, $Z\gamma$, $\gamma\gamma$, gg	I12	p-wave annihilation (sub-dominant); condensate mass shifts; field-dependent gauge couplings
15	radion couplings	σ with all massive fields	$\sigma\, m\,\frac{\partial \ln m}{\partial \ln f}\,\bar{\psi}\psi$, etc.	the masses track the block size
16	Ψ (x_0-chiral) - baryon current	massive states of Ψ, the six quarks	I13	s-wave annihilation: the dark-matter relic; direct detection

6 The x_0 mass mechanism

The author's requirement: mass is given to every non-massless particle by the dependence of its wave function on the space coordinate x_0, periodic with period $\pi/(6H)$; localisation in x_0 implies, through $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$, a hidden momentum and hence an effective mass. UM-2 implements it exactly: every four-dimensional mass is an eigenvalue of the x_0 operator of its field, acting on wave functions of $y(z)=\ln|\sin z|/(6H)$, $z=6Hx_0$, which are periodic in x_0 with period $\pi/(6H)$. Throughout this section the block is static (f constant), $\xi=y/L\in[-1,0]$ on each half, and eigenvalues are given in units of the proper half cell, $\Lambda=(m\,fL)^2$ for bosons and $e=\varepsilon\,fL$ for fermions.

6.1 Gauge fields and the graviton (DERIVED)

A four-dimensional component $A_\mu(x)\,u(y)$ of a gauge field (gauge $A_y=0$) has the reduced action $\int_{-L}^{0}dy\,f^4e^{3Hy}\left[(\partial_y u)^2/f^2-m^2u^2\right]$ per half, hence the x_0 operator and boundary conditions

$$-\frac{1}{f^2}\,e^{-3Hy}\,\partial_y\left(e^{3Hy}\partial_y u\right)=m^2u\quad\Longleftrightarrow\quad-e^{-3\eta\xi}\left(e^{3\eta\xi}u'\right)'=\Lambda\,u,$$

$$\text{fold: } u'(0)=0\ \text{(even)},\quad u(0)=0\ \text{(odd)};\qquad \text{tip: } u'(-1)=\beta\,u(-1),$$

with $\beta = 0$ for the unbroken generators (photon, gluons) and $\beta > 0$ for the broken ones ($W^{\pm}$, Z; from the Σ term). The transverse-traceless graviton obeys the same operator in Einstein gravity. With $u = e^{-3\eta\xi/2}\tilde{u}$ and $q^2 = \Lambda - 9\eta^2/4$ (primes denote $d/d\xi$) the solutions are elementary and the eigenvalue conditions are

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta\, q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0$$

($q = i\tilde{q}$ below the gap $9\eta^2/4$).

Massless particles. For $\beta = 0$ the flat profile $u = \text{const}$ is an exact eigenfunction with $\Lambda = 0$: the **graviton, the photon and the gluons are exactly massless**; their masslessness is protected by four-dimensional general covariance and gauge invariance. The excited levels are $\Lambda_n = 9\eta^2/4 + n^2\pi^2$ (even, $n \geq 1$) and the roots of $q\cot q = -3\eta/2$ (odd); for $\eta = 3$: $\Lambda_1 = 30.1196$, odd 27.0890, gap 20.25 (DERIVED, finite elements agree to 5×10^{-10}). The Kaluza-Klein scale is $m_{KK} = \sqrt{\Lambda_1}/(fL)$.

$W^{\pm}$ and Z. The Σ term gives β_W for $W^{\pm}$ (one value for both charges) and, by custodial symmetry, $\beta_Z = \beta_W/\cos^2\theta_W$, with θ_W the angle of the gauge couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2+\hat{g}'^2)$; at small β, $\Lambda \simeq \beta\cdot 3\eta e^{-3\eta}/(1-e^{-3\eta})$ and $m_Z/m_W = 1/\cos\theta_W$. Beyond leading order the bending of the profiles moves the mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ (Λ_W, Λ_Z the lowest even eigenvalues for β_W and β_Z) away from $\cos^2\theta_W$: the electroweak parameter

$$\rho = \frac{\Lambda_W}{\Lambda_Z\cos^2\theta_W}$$

exceeds 1. This tree-level custodial violation is what the measured $\rho_0 - 1 = 0.00038$ constrains (ρ_0 measures the violation beyond the Standard-Model loops). With the measured masses (STANDARD: $m_W = 80.3625$ GeV, $m_Z = 91.1879$ GeV, PDG 2026; $\rho_0 - 1 = 0.00038$), $\rho = m_W^2/(m_Z^2\cos^2\theta_W) = \rho_0$ fixes the angle of the couplings, $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (the on-shell value m_W^2/m_Z^2 is 0.776663), and the fit gives (DERIVED)

$$\beta_W = 0.011912, \qquad \Lambda_W = 1.32149 \times 10^{-5}, \qquad m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W} = 121.3 \text{ TeV},$$

with both masses reproduced: $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W} = 91.1879$ GeV. A positive tip term cannot make $W^{\pm}$ heavier than the Dirichlet limit $\tanh\tilde{q} = 2\tilde{q}/(3\eta)$, $\Lambda_D = 0.010016$, i.e. $m_W \leq 0.01824\, m_{KK}$.

6.2 Fermions (DERIVED)

The eight-dimensional Dirac equation with the canonical spin connection reduces EXACTLY (all eight blocks, general $a_4(x_4)$, $b_4(x_4)$, y-dependent mass $M(y)$ and potential $v(y)$; DERIVED with the author's matrices) under

$$\Psi = e^{ikx_1}\, e^{-\frac{3}{2}Hy}\, e^{a_4/2}\, b_4^{-1}\, V_\beta\, \chi(y, x_4)$$

to $i\partial_4\chi = h_j\chi$ (k the momentum of the mode along x_1) with the 2×2 block Hamiltonian

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big] + v(y),$$

where V_β ($\beta = 0..7$) are the columns of the block basis of the commuting operators $J = \gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$, $K_1 = \gamma^{x_2}\gamma^{x_3}$, $K_2 = \gamma^{x_5}\gamma^{x_6}$ with eigenvalues (j, is_2, is_3). In every block $\gamma^{x_0} = \sigma_3$, $B = js_2$, $C = s_2\sigma_2$ and $\Gamma_4 = -js_2\sigma_3$. The factor $e^{-\frac{3}{2}Hy}$ removes the $\frac{3}{2}(H/f)\gamma^{x_0}$ of the spin connection, and $e^{a_4/2}b_4^{-1}$ removes its γ^{x_4} part. For a constant mass μ and $k = 0$ the general solution of $h_j\chi = 0$ is $\chi = (A\,e^{f\mu y}, B\,e^{-f\mu y})$; the boundary conditions of section 5.5 select one component. With $c = f\mu L$:

- doublet zero mode $\chi = (e^{c\xi}, 0)$, $\Gamma_4 = +1$, localised at the fold for $c > 0$;
- singlet zero mode $\chi = (0, e^{-c\xi})$, $\Gamma_4 = -1$, localised at the fold for $c < 0$;
- at $k \neq 0$: $h_j(e^{c\xi}, 0) = je^{-a_4}k\,(e^{c\xi}, 0)$ exactly: the zero modes are massless four-dimensional particles with $\varepsilon = e^{-a_4}|k|$ (physical momentum).

Masses from the tip: one generation. The tip coupling turns the tip conditions into the self-adjoint mixing condition (DERIVED)

$$b_L(-1) = \kappa_f\, b_R(-1), \qquad a_R(-1) = -\kappa_f\, a_L(-1), \qquad \chi = (a, ib),$$

with κ_f the dimensionless tip mixing. At $k=0$ the doublet ($c_L>0$) and singlet ($c_R<0$) solutions of $a'=c\,a-e\,b$, $b'=e\,a-c\,b$ (block $j=+1$, eigenvalue $e=\varepsilon f L$) that satisfy the fold conditions reach the tip with

$$(a_L,b_L)=(C_L-c_L s_L,\ -s_L e),\qquad (a_R,b_R)=(s_R e,\ C_R+c_R s_R),\qquad s=\frac{\sinh w}{w},\quad C=\cosh w,\quad w=\sqrt{c^2-e^2},$$

so the four-dimensional mass is the lowest root of the exact x_0 eigenvalue equation (DERIVED; direct ODE shooting of the coupled system agrees to 6×10^{-10})

$$D(e)=e^2 s_L s_R-\kappa_f^2\,(C_L-c_L s_L)(C_R+c_R s_R)=0,$$

whose first-order solution is

$$e=\kappa_f\,\zeta(c_L)\,\zeta(-c_R),\qquad \zeta(c)^2=\frac{c\,e^{-c}}{\sinh c}\simeq 2c\,e^{-2c}\quad (c>0),$$

i.e. the tip mixing times the probability densities of the two chiral wave functions at the tip ($c_L>0$ and $c_R<0$: both zero modes are localised at the fold).

Masses and mixing from the tip: three generations. In each sector (neutrinos, charged leptons, up-type quarks, down-type quarks) the tip coupling is a 3×3 matrix κ (the dimensionless form of λ_ν, λ_e, λ_u, λ_d), and the conditions hold for the vectors of boundary values over the generations:

$$b_L(-1)=\kappa\,b_R(-1),\qquad a_R(-1)=-\kappa^\dagger a_L(-1),$$

which is self-adjoint for every complex κ (DERIVED: the boundary form vanishes identically, sympy, 42 real variables). With the doublet amplitudes $\mathcal{A}_\alpha$ and the singlet amplitudes $\mathcal{B}_i$ of the solutions above, $\mathrm{diag}(b_{L\alpha})\mathcal{A}=\kappa\,\mathrm{diag}(b_{Ri})\mathcal{B}$ and $\mathrm{diag}(a_{Ri})\mathcal{B}=-\kappa^\dagger\mathrm{diag}(a_{L\alpha})\mathcal{A}$, hence

$$\Big[\mathrm{diag}(b_{L\alpha})+\kappa\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\,\mathrm{diag}(a_{L\alpha})\Big]\mathcal{A}=0:$$

the masses are the roots e of the determinant of this matrix, and $\mathcal{A}$ is the left-handed flavour content of each mass state. If the three doublets of a sector share one bulk mass ($a_{L\alpha}=a_L$, $b_{L\alpha}=b_L$) and κ is unitary, the matrix is $\kappa\big[b_L+a_L\,\mathrm{diag}(b_{Ri}/a_{Ri})\big]\kappa^\dagger$, so

$$\det\big[\,\ldots\big]=\prod_{i=1}^{3}\frac{b_L a_{Ri}+a_L b_{Ri}}{a_{Ri}},\qquad \mathcal{A}^{(i)}=\kappa\,\hat e_i,\qquad \mathcal{B}^{(i)}\propto\hat e_i$$

($\hat e_i$ the unit vectors): each factor is the one-generation condition with $\kappa_f=1$ ($b_L a_R+a_L b_R=-D(e)$), so the masses are the one-generation roots, the left-handed rotation of the sector is its tip matrix (so that the mixing matrices are $U=\kappa_e^\dagger\kappa_\nu$ and $V=\kappa_u^\dagger\kappa_d$), and the right-handed rotations are trivial (DERIVED with mpmath at 50 digits: the roots of the 3×3 determinant equal the one-generation roots, both refined in the same computation, to 3.0×10^{-42} in all four sectors; the masses stored by the mass computation, found with a stopping tolerance near 10^{-11}, agree with them to 4.5×10^{-11}; the null vectors equal the columns of κ to 1.1×10^{-16}; for a random complex non-unitary κ the exact roots agree with the singular values of $\zeta(c_L)\,\kappa\,\mathrm{diag}(\zeta(-c_{Ri}))$ to 3.5×10^{-5}).

The UM-2 choice (ASSUMED). One doublet bulk mass per sector, common to the three generations ($c_L=f\mu_L L$ for the leptons, $c_Q=f\mu_Q L$ for the quarks), fixed by the heaviest fermion of the sector with $c_R=-c_L$ (τ) and $c_R=-c_Q$ (t); unitary tip matrices

$$\kappa_e=\kappa_u=1,\qquad \kappa_\nu=U_{PMNS},\qquad \kappa_d=V_{CKM}$$

in the basis in which the singlet bulk masses are diagonal. The measured masses (STANDARD, the PDG 2025 values: m_e, m_μ, m_τ; m_u, m_d, m_s in $\overline{MS}$ at 2 GeV, $m_c(m_c)$, $m_b(m_b)$, the directly measured m_t, used as tree-level targets) and the neutrino masses of section 10.1 then require (DERIVED; exact roots; the first-order formula agrees to 1.1×10^{-3} for the top quark and better than 3×10^{-4} for every other fermion)

fermion	mass	doublet c	singlet c_R	μ_R/H	singlet width $fL/(2\lvert c_R\rvert)$	$e = m\,fL$
ν_1	1.000 meV	5.9529	-35.0452	-54.194	1.27×10^{-22} m	4.524×10^{-17}
ν_2	8.712 meV	5.9529	-32.8481	-50.796	1.36×10^{-22} m	3.941×10^{-16}
ν_3	50.140 meV	5.9529	-31.0702	-48.047	1.44×10^{-22} m	2.268×10^{-15}
e	0.51100 MeV	5.9529	-14.5539	-22.506	3.07×10^{-22} m	2.312×10^{-8}
μ	105.658 MeV	5.9529	-8.9809	-13.888	4.97×10^{-22} m	4.779×10^{-6}
τ	1776.93 MeV	5.9529	-5.9529	-9.205	7.50×10^{-22} m	8.038×10^{-5}
u	2.16 MeV	3.3823	-15.4300	-23.861	2.89×10^{-22} m	9.771×10^{-8}
c	1.2730 GeV	3.3823	-8.7683	-13.559	5.09×10^{-22} m	5.758×10^{-5}
t	172.56 GeV	3.3823	-3.3823	-5.230	1.32×10^{-21} m	7.806×10^{-3}
d	4.70 MeV	3.3823	-14.6258	-22.617	3.05×10^{-22} m	2.126×10^{-7}
s	93.5 MeV	3.3823	-11.5158	-17.808	3.88×10^{-22} m	4.230×10^{-6}
b	4.183 GeV	3.3823	-7.5005	-11.599	5.95×10^{-22} m	1.892×10^{-4}

The doublet bulk masses are $\mu_L = 9.2055\,H$ (width 7.50×10^{-22} m) and $\mu_Q = 5.2304\,H$ (width 1.32×10^{-21} m). Order-one bulk masses give the whole hierarchy, because the tip densities depend exponentially on c; within a sector the lighter fermion has the sharper singlet localisation. The neutrinos are Dirac fermions; their right-handed partners ν_{Ri} are gauge singlets that are never thermalised. (Version 1 gave every fermion its own $c_L = -c_R$; with three generations that rule would give the members of one doublet different bulk masses, against the single bulk mass of a multiplet, and flavour-changing Kaluza-Klein couplings, below.)

Mixing matrices (STANDARD parametrisation, measured values). Both tip matrices are written as

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}, \qquad J = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta,$$

with, for the PMNS matrix (NuFIT 6.0, normal ordering), $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta = 212^\circ$ ($\theta_{12} = 33.71^\circ$, $\theta_{23} = 43.28^\circ$, $\theta_{13} = 8.56^\circ$; $J = -0.01777$), and for the CKM matrix (PDG global fit) $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ ($J = 3.117\times10^{-5}$). Both are unitary to 1.1×10^{-16} (DERIVED). The magnitudes are

$$|U| = \begin{pmatrix} 0.8226 & 0.5488 & 0.1488 \\ 0.3351 & 0.6543 & 0.6779 \\ 0.4594 & 0.5203 & 0.7199 \end{pmatrix}, \qquad |V| = \begin{pmatrix} 0.97435 & 0.22501 & 0.00373 \\ 0.22488 & 0.97349 & 0.04183 \\ 0.00858 & 0.04111 & 0.99912 \end{pmatrix}.$$

No flavour-changing neutral currents at tree level (DERIVED). The first even Kaluza-Klein gauge mode couples to a zero mode of localisation c through the overlap $\int|\chi_c|^2 u_1\,d\xi$ (u_1 normalised like the zero mode); relative to the zero mode this is 0.14434 for every left-handed lepton and -0.30066 for every left-handed quark, because the doublets share one bulk mass, and the right-handed rotations are trivial, so in the mass basis the largest off-diagonal

coupling is 10^{-17} (round-off). With different doublet bulk masses the PMNS rotation would create off-diagonal couplings up to 0.033. The W and Z zero modes couple to every generation with the same strength (lepton universality); the PMNS and CKM matrices carry the whole flavour structure of the charged currents. Charged-lepton flavour violation through the neutrino masses, $\mathrm{BR}(\mu\to e\gamma)=\frac{3\alpha}{32\pi}\big|\sum_i U^*_{\mu i}U_{ei}\,m_i^2/m_W^2\big|^2=2.8\times10^{-55}$, is unobservable.

6.3 The quanta of dirac16complex (DERIVED)

Ψ has the same block Hamiltonian with $M=m$ and the boundary conditions fold $(1-\gamma^{x_0})\Psi=0$ (even) or $(1+\gamma^{x_0})\Psi=0$ (odd) and tip $(1-\gamma^{x_0})\Psi=0$; both parities are modes of the full period. With $c_\psi=fmL$ the exact $k=0$ spectrum is

- even: the zero mode $\chi=(e^{c_\psi\xi},0)$, $\varepsilon=0$, in all eight blocks, and $\varepsilon=\pm\sqrt{m^2+(n\pi/(fL))^2}$;
- odd: $\varepsilon=\pm\sqrt{m^2+p^2}$ with $\tan(p\,fL)=-p/m$.

The **fold band** $\varepsilon=e^{-a_4}|k|$ is massless: 8 states per 3-momentum, 4 in Ψ_A (two Weyl species with $\Gamma_4=+1$) and 4 in Ψ_B (two Weyl species with $\Gamma_4=-1$). The **bulk band** is massive; at the stabilised block ($f_*=0.21556$, $c_\psi=0.64667$) its lowest level is the odd level $\varepsilon_{DM}=3.10220\,m$ ($p\,fL=1.89901$, $p=2.93661\,m$), and the lowest even bulk level is $4.95998\,m$. The odd level is the dark-matter candidate (section 11).

6.4 What the uncertainty relation does and does not bound (DERIVED)

In the flat measure $u=e^{-3\eta\xi/2}\tilde u$ the exact identity $e^{3\eta\xi}u'^2=\tilde u'^2+\frac94\eta^2\tilde u^2-\frac32\eta\,(\tilde u^2)'$ turns every bosonic eigenvalue into

$$(m\,fL)^2=\langle p^2\rangle+\frac{9\eta^2}{4}-\frac{3\eta}{2}\,\varrho(\mathrm{fold})+\Big(\frac{3\eta}{2}+\beta\Big)\varrho(\mathrm{tip}),\qquad \langle p^2\rangle=\frac{\int\tilde u'^2}{\int\tilde u^2},\quad \varrho=\frac{\tilde u^2}{\int\tilde u^2},$$

and every fermionic one into $(\varepsilon fL)^2=\langle|\chi'|^2\rangle+c^2-c\,[\chi^\dagger\sigma_3\chi]_{-1}^0/\int|\chi|^2$, with $\langle p^2\rangle=\langle|\chi'|^2\rangle$ and $\varrho=|\chi|^2/\int|\chi|^2$ (integrals over $\xi\in[-1,0]$, primes $d/d\xi$). The proper uncertainty relation bounds only the hidden kinetic energy. On the cell, which ends at the tips, it reads

$$\sqrt{\langle p^2\rangle}\,\Delta\xi\ge\frac{1-\varrho(\mathrm{tip})}{2},\qquad \Delta\xi^2=\int_{-1}^{0}\xi^2\varrho\,d\xi,$$

since $1=\int\varrho\,d\xi=\varrho(\mathrm{tip})-\int\xi\varrho'\,d\xi$ (the fold end $\xi=0$ contributes nothing) and the Cauchy-Schwarz inequality bounds $|\int\xi\varrho'\,d\xi|$ by $2\Delta\xi\sqrt{\langle p^2\rangle}$; $\Delta s=fL\,\Delta\xi$ is the root-mean-square proper distance from the fold. The familiar $\langle p^2\rangle\ge1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip (it can hold otherwise: the first even Kaluza-Klein mode, with $\varrho(\mathrm{tip})=0.655$, has $\langle p^2\rangle\Delta\xi^2=4.59$); the photon zero mode has 0.497, the broad fold zero mode of Ψ ($\varrho(\mathrm{tip})=0.489$) only 0.097. The relation with the tip term is a theorem; it is also checked numerically for each of the 21 computed states (the zero mode of the photon, gluons and graviton, $W^\pm$, Z, the first even and odd Kaluza-Klein modes, the two doublet zero modes c_L and c_Q, the twelve singlet zero modes of the fermions, the fold zero mode of Ψ and the dark matter; DERIVED). Three classes follow:

- **massless**: profiles flat relative to the geometry ($u=$ const or $h\chi=0$): the fold term cancels the kinetic energy and the warp exactly; graviton, photon, gluons and the Ψ fold band. They are localised at the fold ($\Delta s=1.4\times10^{-21}$ m for the photon) and yet massless;
- **heavy**: the x_0 dependence fills the cell. For the gauge fields and the graviton every odd mode and every excitation has $m\ge3H/(2f)=99.5$ TeV (the gap theorem of the boson operator: one even level below the gap for every $\beta\ge0$, no odd level); the Ψ excitations start at its lowest odd level, $3.10m=44.35$ TeV (the dark matter), since the Dirac operator has no warp gap. For the computed states of this class the author's estimate $\hbar c/(2\Delta s)$, with Δs the root-mean-square proper distance from the fold, lies below the mass (by factors 2.3 to 7.5, the dark matter included); this is observed for these states, not derived for all;
- **light**: the x_0 dependence that costs mass sits at the tip, where the probability is tiny: $W^\pm$, Z and the twelve fermions. Their masses lie 3×10^{-3} (the top quark) to 10^{-17} (ν_1) below $\hbar c/(2\Delta s)$ (Δs of the broader of the fermion's two zero modes; 10^{-3} for $W^\pm$), and a sharper localisation of either zero mode gives a LIGHTER fermion ($d\ln m_f/d\ln c=-2.9$ to -34.5).

7 Dimensional reduction and the four-dimensional constants

The good sector (no dependence on x_5, x_6, x_7) is kept (ASSUMED): modes with extra-time momentum q have $\omega^2 = M^2 + k^2 - q^2/f^2$ and are tachyonic for $q > fM$ (OPEN). The extra times are compact with coordinate volume v_t (ASSUMED). Integrating over one period of x_0 (both halves) and the extra times, the proper block volume per unit 4-volume is

$$V_\perp = 2\int_{-L}^{0} dy\, f^4 e^{3Hy}\, v_t = \frac{2f^4 v_t\,(1-e^{-3\eta})}{3H},$$

and the reduced action of section 8.5 gives

$$\bar{M}_{Pl}^2 = \frac{G(k)\,V_\perp}{\kappa} = \frac{1}{8\pi G_N}, \qquad g^2 = \frac{\hat{g}^2}{V_\perp}, \quad g'^2 = \frac{\hat{g}'^2}{V_\perp}, \quad g_s^2 = \frac{\hat{g}_s^2}{V_\perp}.$$

($\bar{M}_{Pl}$ is the reduced Planck mass). Every four-dimensional constant depends on the block: α and the gauge couplings $\propto f^{-4}$, $G_N \propto f^{+2.05}$ ($d\ln G_N/d\ln f = -4 + 2kG'(k)/G(k)$ at k_*: the Lovelock factor varies too), Kaluza-Klein masses $\propto f^{-1}$, $m_W \propto f^{-0.50}$, the fermion masses as $f^{-6.8}$ (top quark) to f^{-41} (ν_1) ($m_e \propto f^{-20.5}$), $M_{DM} \propto f^{-0.77}$ (DERIVED). The masses are constant only because the block is stabilised.

8 Gravity: the background, the stabilised block, the deflating extra times

8.1 The background field equations (DERIVED)

For the order-0 metric every off-diagonal component of every $E_{(k)}$ vanishes and the hidden and extra-time components coincide, so the source must be diagonal, independent of x_0 and block-isotropic:

$$T^M{}_N = \mathrm{diag}(p_0, p_3, p_3, p_3, -\rho, p_t, p_t, p_t), \qquad p_0 = p_t.$$

The three independent components are the x_4 constraint (first order in time derivatives), the 3-space component and the block component; the last two are linear in $(\dot{A}, \dot{h})$. With the pairing formula of section 5.1, the sectional curvatures of section 4.1, the author's weights and $\Lambda = -10H^2$ they are ($A = \dot{a}_4$, $h = \dot{f}/f$, $k = 1/f^2$, dots d/dx_4; DERIVED, sympy)

$$\begin{aligned}
x_4: \quad & -\frac{15A^2h^4}{16H^4} - \frac{5A^3h^3}{4H^4} - \frac{3h^4}{8H^2} - \frac{9Ah^3}{2H^2} - \frac{27A^2h^2}{4H^2} + \frac{9kA^2h^2}{8H^2} \\
& - \frac{3hA^3}{2H^2} + \frac{3hkA^3}{4H^2} - 6h^2 + \frac{3kh^2}{4} - 12Ah + \frac{9Ahk}{2} \\
& - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 + 6kH^2 - \frac{3H^2k^2}{8} \\
& = -\kappa\rho \\
x_1: \quad & \Big[-\frac{h^4}{8H^4} - \frac{Ah^3}{2H^4} - \frac{3h^2}{2H^2} + \frac{kh^2}{4H^2} - \frac{Ah}{H^2} + \frac{Ahk}{2H^2} \\
& \quad - 2 + \frac{3k}{2} - \frac{k^2}{8} \Big]\dot{A} \\
& + \Big[-\frac{Ah^3}{2H^4} - \frac{3A^2h^2}{4H^4} - \frac{3h^2}{2H^2} - \frac{3Ah}{H^2} + \frac{Ahk}{2H^2} - \frac{A^2}{2H^2} \\
& \quad + \frac{kA^2}{4H^2} - 4 + \frac{3k}{2} \Big]\dot{h} \\
& - \frac{Ah^5}{2H^4} - \frac{19A^2h^4}{16H^4} - \frac{A^3h^3}{2H^4} - \frac{15h^4}{8H^2} - \frac{6Ah^3}{H^2} + \frac{Akh^3}{2H^2} \\
& - \frac{17A^2h^2}{4H^2} + \frac{7kA^2h^2}{8H^2} - \frac{hA^3}{H^2} + \frac{hkA^3}{2H^2} - 10h^2 + \frac{9kh^2}{4} \\
& - 8Ah + 3Ahk - 3A^2 + \frac{9kA^2}{4} - \frac{3A^2k^2}{16} - 10H^2 \\
& + 6kH^2 - \frac{3H^2k^2}{8} \\
& = \kappa p_3
\end{aligned}$$

$$
\begin{aligned}
x_0:\quad & \Big[-\frac{3Ah^3}{8H^4}-\frac{9A^2h^2}{16H^4}-\frac{9h^2}{8H^2}-\frac{9Ah}{4H^2}+\frac{3Ahk}{8H^2}-\frac{3A^2}{8H^2}\\
&\quad+\frac{3kA^2}{16H^2}-3+\frac{9k}{8}\Big]\dot A\\
&+\Big[-\frac{9A^2h^2}{16H^4}-\frac{3hA^3}{8H^4}-\frac{3h^2}{8H^2}-\frac{9Ah}{4H^2}-\frac{9A^2}{8H^2}+\frac{3kA^2}{16H^2}\\
&\quad-3+\frac{3k}{8}\Big]\dot h\\
&-\frac{9A^2h^4}{16H^4}-\frac{17A^3h^3}{16H^4}-\frac{9A^4h^2}{16H^4}-\frac{3h^4}{8H^2}-\frac{27Ah^3}{8H^2}-\frac{45A^2h^2}{8H^2}\\
&+\frac{3kA^2h^2}{16H^2}-\frac{27hA^3}{8H^2}+\frac{9hkA^3}{16H^2}-\frac{3A^4}{8H^2}+\frac{3kA^4}{16H^2}-6h^2\\
&+\frac{3kh^2}{8}-9Ah+\frac{9Ahk}{8}-6A^2+\frac{9kA^2}{4}-10H^2\\
&+3kH^2\\
&=\kappa\bar p
\end{aligned}
$$

with $\bar p=(p_0+3p_t)/4$, and the source is conserved,

$$\dot\rho=-3A(\rho+p_3)-h(\rho+p_0)-3h(\rho+p_t)=-3A(\rho+p_3)-4h(\rho+\bar p).$$

A source with $p_0\neq p_t$ (the Kohn-Sham states of Ψ and the matter at the fold have it; the dark matter has $p_0/\rho=0.772$, $p_t=0$) acts on the common block scale f through the block-averaged pressure $\bar p=(p_0+3p_t)/4$ (the variation of the action with respect to f), while p_0-p_t drives a block-shape mode that the order-0 metric does not contain (OPEN); for matter its size relative to the stabiliser is of order $(A/H)^2$, 10^{-92} today. To first order about the static block of section 8.2 with block-averaged matter sources (A^2 counts as first order: the x_4 constraint gives $3G(k_*)A^2=\kappa\delta\rho$, because $\frac{d}{dk}[6H^2k(1-2w_2k)+\kappa\rho_b]$ vanishes at k_*), the equations give (DERIVED)

$$\delta\dot A=J_{Ak}\,\delta k+s_{A3}\,\kappa\delta p_3+s_{Ap}\,\kappa\delta\bar p+q_A\,\kappa\delta\rho,$$

$$\delta\dot h=J_{hk}\,\delta k+s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho,\qquad \delta\dot k=-2k_*\,\delta h,$$

$$\delta\ddot k+\omega_r^2\,\delta k=-2k_*\big(s_{h3}\,\kappa\delta p_3+s_{hp}\,\kappa\delta\bar p+q_h\,\kappa\delta\rho\big),\qquad \omega_r^2=2k_*J_{hk},$$

with $J_{Ak}=\frac{6(238-19\sqrt{51})}{2249}H^2=0.272956H^2$, $J_{hk}=\frac{4063-485\sqrt{51}}{2249}H^2=0.266522H^2$, $s_{A3}=\frac{2283}{4498}-\frac{162\sqrt{51}}{2249}=-0.006853$, $s_{Ap}=\frac{28\sqrt{51}-114}{2249}=0.038221$, $q_A=\frac{135021}{265382}-\frac{10308\sqrt{51}}{132691}=-0.045997$, $s_{h3}=\frac{21\sqrt{51}}{2249}-\frac{171}{4498}=0.028666$, $s_{hp}=\frac{239\sqrt{51}}{6747}-\frac{485}{2249}=0.037320$, $q_h=\frac{57}{4498}-\frac{7\sqrt{51}}{2249}=-0.009555$. In the adiabatic response ($\delta\ddot k=0$) the block shifts by $\delta k=(-0.107556\,\kappa\delta p_3-0.140028\,\kappa\delta\bar p+0.035852\,\kappa\delta\rho)/H^2$ and 3-space obeys exactly $\delta\dot A=-\kappa(\delta\rho+\delta p_3)/(2G(k_*))$: the four-dimensional Friedmann acceleration equation with $8\pi G_N=\kappa/(G(k_*)V_\perp)$.

8.2 The static block and attractive gravity with the author's weights (DERIVED)

For a static block ($h=0$, k constant) and $\dot A=0$ the components are

$$x_4:\quad -3G(k)A^2+6H^2k(1-2w_2k)+\Lambda+\kappa\rho=0,\qquad x_1:\quad \text{the same with } -\kappa p_3 \text{ for } \kappa\rho,$$

$$x_0:\quad 6(12w_2k-1)A^2-\frac{12(w_2-18w_3k)}{H^2}A^4+3H^2k+\Lambda-\kappa p_0=0,\qquad G(k)=1-24w_2k+72w_3k^2.$$

$G(k)$ is the coefficient of the four-dimensional Einstein-Hilbert term (section 8.5): gravity is attractive for $G>0$. Without sources the only static block with $A=0$ (and $k\neq0$) is $k_0=1/(4w_2)=8$ with $\Lambda_0=-24H^2$ and $G(k_0)=-5+9w_3/(2w_2^2)=-1$: with the author's weights the sourceless block has REPULSIVE gravity, and the author's $\Lambda=-10H^2$ is not Λ_0. For the author's weights $G(k)=(k^2-12k+16)/16$, positive only for $k<6-2\sqrt5$ or $k>6+2\sqrt5=10.47$.

With the block stabiliser ($p_3=-\rho_b$, $p_0=p_t=\rho_b$) and $A=0$ the two conditions require

$$\kappa\rho_b=10H^2-6H^2k+\tfrac38H^2k^2=\kappa p_0=3H^2k-10H^2\quad\Longrightarrow\quad k_\pm=12\pm\tfrac43\sqrt{51}.$$

The upper root is the UM-2 vacuum:

$$k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219, \qquad f_* = k_*^{-1/2} = 0.215556,$$

$$G(k_*) = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0, \qquad \kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2,$$

with attractive four-dimensional gravity and the author's weights and Λ unchanged. The lower root $k_- = 2.4781$ has $G = -0.4748$.

8.3 Stability: the radion (DERIVED)

Linearising the 3-space and block components (solved for $\dot{A}$, $\dot{h}$) with $\dot{k} = -2kh$ and $\rho_b \propto k^4$ about k_* gives the eigenvalues $0, \pm i\,\omega_r$ with

$$\frac{\omega_r^2}{H^2} = \frac{31552}{2249} - \frac{2416\sqrt{51}}{6747} = 11.4721, \qquad \omega_r = 3.38705\,H :$$

the block is stable and oscillates; the zero eigenvalue is the direction of a small four-dimensional cosmological term. The radion (the quantum of block-size oscillations) has the mass $m_r = 3.387H = 48.4$ TeV. Stability holds for each of the stabiliser laws $p_0 = \nu\rho_b$, $\rho_b \propto k^{2(1+\nu)}$, computed at $\nu = 1/4, 1/2, 3/4, 1, 5/4, 3/2, 2, 3$ (radion masses $3.33H$ to $4.10H$).

8.4 The deflating extra times (DERIVED, full nonlinear equations)

The full nonlinear equations (all three components, the x_4 constraint preserved to 6×10^{-10}) were integrated from an expanding 3-space (A from the constraint) and a block LARGER than the stabilised one ($k_0 < k_*$, i.e. $f_0 > f_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for k_0 from 1.5 to 20. Every start with k_0 above the separatrix $k_c(h_0) = 2.6983, 2.5991, 2.5301$ (for $h_0 = -H/2, -H/3, -H/10$; just above the unstable static block $k_- = 2.4781$, whose stable manifold the separatrix is) ends at k_*: x_5, x_6, x_7 and x_0 deflate (at rates up to $|h| = 2.63H$ in the stored runs) by up to the factor $f(k_c)/f_* = (k_*/k_c)^{1/2} = 2.917$, reach k_* for the first time within $x_4 < 1/H$, and then oscillate about k_* with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 = 0.667$ to 0.670 late, $2/3$ for dust). Starts with $k_0 < k_c$ (still larger blocks) become singular within $x_4 = 0.2/H$ to $1.4/H$ or admit no expanding initial data. The extra times therefore deflate exponentially only for a short primordial era and are then frozen, which is what the constancy of the masses requires (section 7).

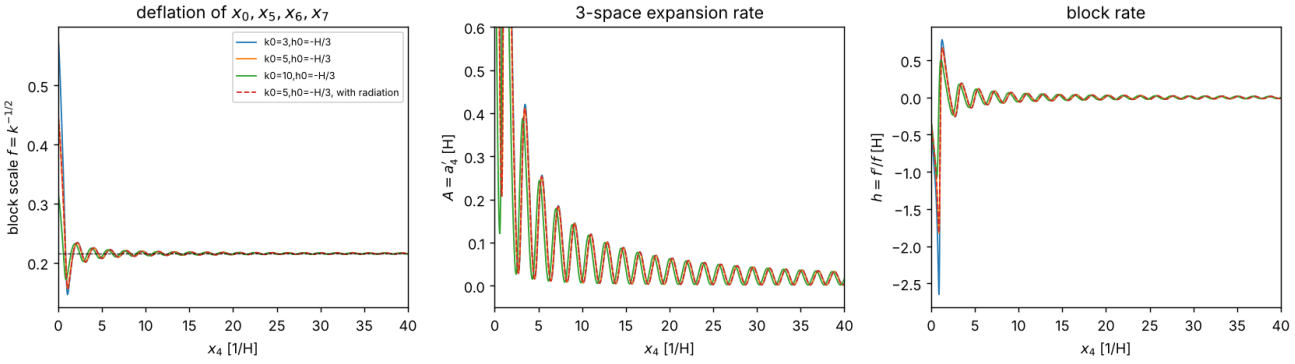


Figure 1: Deflation of the extra times into the stabilised block (full nonlinear Einstein-Lovelock equations with the author's weights).

8.5 The four-dimensional effective gravity (DERIVED)

On a product of a four-dimensional metric with the static block (constant curvature $K = -H^2k$), the Lovelock densities factorise, $\mathcal{L}_{(p)} = \sum_{r+s=p} \frac{p!}{r!\,s!}\mathcal{L}^{(4)}_{(r)}\mathcal{L}^{(int)}_{(s)}$ ($p = 1, 2, 3$ the Lovelock order) with $\mathcal{L}^{(int)}_{(s)} = (1, 12K, 24K^2, 0)$ for

$s = 0, 1, 2, 3$, and $\mathcal{L}^{(4)}_{(3)} \equiv 0$ in four dimensions while $\mathcal{L}^{(4)}_{(2)}$ (the four-dimensional Gauss-Bonnet density) is topological. The reduced Lagrangian (gravity plus the stabiliser, in units of $1/(2\kappa)$) is

$$\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)\, R^{(4)} + (\text{topological}) - 2\, G(k)\, \Lambda_4, \qquad \Lambda_4 = \frac{6H^2 k(1-2w_2 k) + \Lambda + \kappa\rho_b}{G(k)},$$

so the four-dimensional gravity of UM-2 is EXACTLY general relativity with Newton's constant $8\pi G_N = \kappa/(G(k_*)V_\perp)$ and the cosmological term Λ_4, which vanishes at the static point by construction. The observed dark energy is a small residual of this balance, $\Lambda_4/H^2 \sim 10^{-92}$: tuned (the cosmological-constant problem, OPEN).

9 dirac16complex in the cell: Kohn-Sham description

Ψ is quantised canonically with x_4 as the time, $\{\Psi_r(x), \Psi^\dagger_s(x')\} = B_{rs}\, \delta^7(x-x')/\sqrt{|g|}$ on a slice (r, s spinor indices); the indefinite Krein metric B is handled in the good sector by counting the positive branch of every block as particles (the record's convention, ASSUMED). The interacting many-body state is described by Kohn-Sham theory: orbitals of h_j with the mean fields

$$M_{eff}(y) = m + \tfrac{15}{16}\lambda S(y) + \delta M_{portal}(y), \qquad v(y) = -\tfrac{1}{16}\lambda\, n(y) + v_{portal}(y),$$

$$e_{int} = \tfrac{15}{32}\lambda S^2 - \tfrac{1}{32}\lambda n^2, \qquad E_{KS} = \sum_o g f_o \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} e^{3Hy} e_{int}\, dy,$$

where n and $S = \bar\Psi\Psi$ are the proper densities, $g = 4r_3(|\vec n|^2)$ the degeneracy of the momentum shell $\vec k = \Delta k\, \vec n$ ($\vec n \in \mathbb{Z}^3$, $r_3(|\vec n|^2)$ the number of representations of $|\vec n|^2$ as a sum of three squares, $\Delta k = 0.25H$), $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f^4$ (at $a_4 = 0$) the normalisation volume of a cell, of dimension length^6 (the proper seven-volume of a cell, with its hidden extent, is $2\,\mathrm{Vol}_7(1 - e^{-3\eta})/(3H)$), f_o the occupations (aufbau or Mermin), and the portal mean fields come from I11 (the densities of the fermions of the three generations), I12 (the gauge condensates) and I13, which acts on the lower component of the orbitals only:

$$h_j \;\to\; h_j + P_b\, v_b(y), \qquad P_b = \tfrac{1}{2}(1 - \sigma_3), \qquad v_b(y) = \frac{c_A}{\Lambda_\star^6}\,\frac{1}{3}\sum_q n_q(y),$$

$$v_{portal}(y) = \frac{c_V}{\Lambda_\star^6}\sum_{i,\psi} q_\psi\, n_{\psi i}(y), \qquad \delta M_{portal} = \frac{\langle \mathcal{F} \rangle}{\Lambda_\star^3},$$

with n_q, $n_{\psi i}$ the proper eight-dimensional densities of the quarks and of the fermions and $\langle \mathcal{F} \rangle$ the condensate of $\mathcal{F} = c_G G^\alpha_{MN} G^{\alpha MN} + c_W W^a_{MN} W^{aMN} + c_B B_{MN} B^{MN}$. The energy-momentum tensor of a Kohn-Sham state,

$$\rho = \sum_o w g f_o \varepsilon_o\, n_o - e_{int}, \qquad p_3 = \sum_o w g f_o\, \tfrac{e^{-a_4|k|}}{3} t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum_o w g f_o \big[(\varepsilon_o - v) n_o - M_{eff} s_o - e^{-a_4}|k| t_o\big] + e_{int},$$

with $n_o = P\chi^\dagger_o \chi_o$, $s_o = P\chi^\dagger_o j\sigma_2 \chi_o$, $t_o = P\chi^\dagger_o j\sigma_3 \chi_o$, $P = e^{-3Hy}/\mathrm{Vol}_7$ and $w = 1/2$ (the densities are $n = \sum_o w g f_o n_o$ and $S = \sum_o w g f_o s_o$), is the source of Ψ in the gravitational equations; its block-averaged pressure $(p_0 + 3p_t)/4$ pushes on the block (section 8.1).

9.1 The coupled solution (DERIVED, version 3)

The coupled equations are solved by iterating the Kohn-Sham equations of Ψ with the mean fields of the Standard-Model fields and the response of the block until self-consistency (16 checks). The four-dimensional couplings of a state of Ψ to the Standard-Model fermions are the overlaps $\tilde I_{of}$ (I11) and $\tilde I^{(b)}_{of}$ (I13) of section 5.7:

zero mode f	Q	L	u_R	c_R	t_R	d_R	s_R	b_R	e_R	μ_R	τ_R	ν_{Ri}
$\tilde{I}$, dark matter	8.272	1.252	0.385	0.606	8.272	0.396	0.462	0.762	0.397	0.588	1.252	0.302-0.308
$\tilde{I}^{(b)}$, dark matter	2.613	0.913	0.379	0.557	2.613	0.389	0.446	0.661	0.390	0.544	0.913	0.301-0.308
$\tilde{I}$, fold band	6.685	1.659	0.792	1.060	6.685	0.807	0.893	1.221	0.808	1.041	1.659	0.668-0.678

(ν_{Ri}: the three right-handed neutrinos, $i = 1, 2, 3$; t_R and τ_R share the profile of their doublets, $c_R = -c_Q$ and $-c_L$; $\tilde{I}^{(b)} = 0$ for the fold band.) Each overlap equals the derivative of the self-consistent Kohn-Sham level with respect to the amplitude of the corresponding mean field to 3×10^{-10} (I11) and 2×10^{-10} (I13; Richardson in the amplitude); in the interacting state (eight fold zero modes, eight dark-matter quanta, λ_B) the response is screened by only 3.5×10^{-4}. The fold band stays massless under every I11 field (its first-order change of the gap is 10^{-10} per unit amplitude, the dispersion stays linear) and is untouched by I13 (to 7×10^{-14}).

The Standard-Model fields acting on Ψ in ten environments. With unpolarised fermions (each chirality half of the density), the net quark densities $n_u = 2n_p + n_n$, $n_d = n_p + 2n_n$ and the electron density n_e, the dark-matter level moves by

$$\delta\varepsilon_{I11} = G_V\Big[\tfrac{1}{3}n_u\tfrac{1}{2}(\tilde{I}_Q + \tilde{I}_{u_R}) + \tfrac{1}{3}n_d\tfrac{1}{2}(\tilde{I}_Q + \tilde{I}_{d_R}) - n_e\tfrac{1}{2}(\tilde{I}_L + \tilde{I}_{e_R})\Big],$$

$$\delta\varepsilon_{I13} = G_A\Big[\tfrac{1}{3}n_u\tfrac{1}{2}(\tilde{I}^{(b)}_Q + \tilde{I}^{(b)}_{u_R}) + \tfrac{1}{3}n_d\tfrac{1}{2}(\tilde{I}^{(b)}_Q + \tilde{I}^{(b)}_{d_R})\Big],$$

$$\delta\varepsilon_{I12} = \mathcal{O}\,\frac{c}{\Lambda_\star^3}\Big[\langle g_s^2 G^\alpha G^\alpha\rangle + \langle g^2 W^a W^a\rangle + \langle g'^2 BB\rangle\Big]$$

(canonical field strengths). The gluon condensate follows from the trace anomaly, $\langle\frac{\alpha_s}{\pi}G^\alpha G^\alpha\rangle_T = \langle\frac{\alpha_s}{\pi}G^\alpha G^\alpha\rangle_0 - \frac{8}{b}(\rho - 3p)_{QCD}$, $b = 11 - \frac{2}{3}n_f$, with $\langle\frac{\alpha_s}{\pi}G^\alpha G^\alpha\rangle_0 = 0.012$ GeV^4 (STANDARD) and $g_s^2 G^2 = 4\pi^2\frac{\alpha_s}{\pi}G^2$; the QCD interaction measure is $(\rho - 3p)/T^4 = 4\exp[-\ln^2(T/0.2\,\text{GeV})/0.5]$ below 1 GeV (a lattice-like shape, ESTIMATE) and $\frac{b}{3}(1 + \frac{5n_f}{12})\alpha_s^2$ above (leading-order perturbation theory, α_s at the scale $2\pi T$, one loop). For the massive vector bosons the free part of the thermal condensate is $\langle F_{\mu\nu}F^{\mu\nu}\rangle_T = -2(\rho - 3p)$ of their ideal gas (on shell the Proca Lagrangian, written here in the signature $(+,-,-,-)$ as $-\frac{1}{4}F^2 + \frac{1}{2}m^2A^2$, is a total derivative, so $\langle -\frac{1}{4}F^2\rangle = -\langle\frac{1}{2}m^2A^2\rangle$, while $T^\mu{}_\mu = -m^2A_\mu A^\mu$; the final relation is independent of the signature convention). Above m_W the interaction pressure of the electroweak plasma dominates: for a coupling g with the Lagrangian $-F_{geo}^2/(4g^2)$ ($F_{geo} = gF$) the Feynman-Hellmann theorem gives

$$\langle F_{geo}^2\rangle = -4\,\frac{\partial p}{\partial(1/g^2)} = 4g^2 p_2, \qquad p_2 = -\frac{g^2T^4}{144}\,d_A\Big[C_A + \frac{5}{2}T_F N_f\Big]$$

for the leading-order pressure $p_2 \propto g^2$ of a gauge plasma with fermions in the massless limit (STANDARD; for QCD, $d_A = 8$, $C_A = 3$, $T_F = \frac{1}{2}$, it reproduces the trace-anomaly value above exactly): for $SU(2)$, $d_A = 3$, $C_A = 2$ and twelve Weyl doublets ($T_F N_f = 3$); for $U(1)_Y$, $d_A = 1$, $C_A = 0$ and $\frac{5}{2}T_F N_f \to \frac{5}{2}\sum_{Dirac}Y^2 = \frac{5}{2}\cdot 5$ ($\sum_{Weyl}Y^2 = 10$ for three generations); no scalar term (ESTIMATE), and below m_W multiplied by the thermal suppression $\rho(m_W, T)/\rho(0, T)$ of a massive vector (ESTIMATE). The couplings are taken at $\mu = 2\pi T$ (one loop from the $\overline{MS}$ values at m_Z, without a Higgs particle: $d(1/\alpha_i)/d\ln\mu = b_i/(2\pi)$, $b_2 = 10/3$, $b_Y = -20/3$). Hence

$$\langle g^2 W^a W^a\rangle = g^2\big[2(-2I_W) + \cos^2\theta_W(-2I_Z)\big] + 4g^2 p_2^{SU(2)}, \qquad \langle g'^2 BB\rangle = g'^2\sin^2\theta_W(-2I_Z) + 4g'^2 p_2^{U(1)},$$

$I_V = (\rho - 3p)_V$; at the freeze-out $\langle g_s^2 G^\alpha G^\alpha\rangle = -8.1 \times 10^{12}$, $\langle g^2 W^a W^a\rangle = -5.0 \times 10^{11}$ (of which -5.0×10^{11} from the interaction) and $\langle g'^2 BB\rangle = -2.6 \times 10^{10}$ GeV^4. Results (DERIVED):

environment	$\delta\varepsilon_{DM}/M_{DM}$	$\delta \ln f$
cosmic mean today; the Galaxy at the Sun; the Earth; the centre of the Sun; a white dwarf	8.4×10^{-20} (the QCD vacuum condensate)	7×10^{-94} to 2×10^{-57}
neutron-star core (0.3 fm^{-3})	1.0×10^{-16}	9×10^{-49}
nucleosynthesis ($T=1$ MeV)	8.4×10^{-20}	7×10^{-60}
QCD crossover ($T=200$ MeV)	4.4×10^{-20}	3×10^{-48}
electroweak era ($T=100$ GeV)	-5.5×10^{-11}	7×10^{-39}
dark-matter freeze-out ($T=1.42$ TeV, $x_f=31.23$)	-1.5×10^{-6}	1.1×10^{-34}

The densities of the environments: the cold ones are given by their matter (cosmic mean today: the Planck baryon density with the helium fraction $Y_p=0.2457$ and $\Omega_c h^2=0.120$; the Galaxy at the Sun: $n_p=n_e=1$ cm^{-3}, $\rho_{DM}=0.4$ GeV/cm^3; the Earth: mean density 5.51 g/cm^3 with 48 % protons; the centre of the Sun: 150 g/cm^3, 66 % protons, 32 % neutrons; a white dwarf: 10^6 g/cm^3 of carbon and oxygen, equal numbers of protons and neutrons; a neutron-star core: 0.3 fm^{-3} with 10 % protons, $n_e=n_p$; the local dark-matter density 0.4 GeV/cm^3 in all bound objects); in the early Universe the net baryon density is $\eta_b n_\gamma$ with $\eta_b=6.1\times10^{-10}$ and $n_\gamma=2\zeta(3)T^3/\pi^2$, shared as $n_n/n_p=e^{-Q/T}$ ($Q=1.293$ MeV), $n_e=n_p$, and the dark matter is in thermal equilibrium, $n=2\cdot4(MT/2\pi)^{3/2}e^{-M/T}$ for both species, while $M/T<40$ (its relic density afterwards).

Ψ acting on the Standard Model. Through I12 the scalar density of the dark matter shifts the gauge couplings, $\delta\alpha_i/\alpha_i=-4G_i n_{DM}$ (at most 1.7×10^{-16} at the freeze-out, 10^{-36} from nucleosynthesis on, 2×10^{-60} today in the Galaxy), and with the kinetic function $1+e\,s_{DM}(y)$ ($s_{DM}=2ab$ the scalar density of the dark-matter orbital) the x_0 eigenvalue of the W changes by $d\ln\Lambda_W/de=-0.1126$ (finite elements), while the photon and the gluons stay exactly massless.

Gravity. The block responds to the block-averaged sources of all species with the adiabatic response of section 8.1,

$$\delta k=\big(0.035852\,\kappa\delta\rho-0.107556\,\kappa\delta p_3-0.140028\,\kappa\delta\bar p\big)/H^2,\qquad \delta\ln f=-\frac{\delta k}{2k_*},$$

in which radiation ($p_3=\rho/3$) drops out exactly ($-0.107556=-3\times0.035852$), so only the trace $\rho-3p_3$ and the hidden pressure $\bar p=(p_0+3p_t)/4$ act; the block averages of four-dimensional densities are $\kappa\delta\rho=8\pi G_N G(k_*)\rho_4$ and so on. The hidden pressure is $\bar p=\frac14\,\partial p/\partial\ln f$ at fixed temperature: a species of mass $m(f)$ contributes $-\frac14(\rho-3p)\,d\ln m/d\ln f$ (non-relativistic matter: $-\frac14\rho\,d\ln m/d\ln f$; the plasma species as ideal gases with their x_0 masses, the nucleons and electrons of the cold environments as degenerate Fermi gases), the QCD plasma $-\frac14(\rho-3p)_{QCD}\,d\ln\Lambda^{(n_f)}/d\ln f$ (for $p=T^4\varphi(T/\Lambda)$, $\partial p/\partial\ln\Lambda=-(\rho-3p)$), and the electroweak plasma $\frac14\sum_i p_2^{(i)}\,d\ln\alpha_i/d\ln f$ with $d\ln\alpha_i(\mu)/d\ln f=-\alpha_i(\mu)\,[4/\alpha_i(m_{KK})+b_i/(2\pi)]$ (the bulk gauge couplings scale as $\alpha_i(m_{KK})\propto f^{-4}$ and $m_{KK}\propto1/f$). The nucleon pushes through the QCD scale: with $\alpha_s(m_{KK})=0.0606$ ($1/\alpha_s(m_{KK})=1/\alpha_s(m_Z)+\frac{7}{2\pi}\ln(m_{KK}/m_Z)$),

$$\frac{d\ln\Lambda^{(6)}}{d\ln f}=-1-\frac{2\pi}{7\alpha_s(m_{KK})}\,4=-60.2,\qquad \frac{d\ln\Lambda^{(3)}}{d\ln f}=\frac{21}{27}\frac{d\ln\Lambda^{(6)}}{d\ln f}+\frac{2}{27}\sum_{q=c,b,t}\frac{d\ln m_q}{d\ln f}=-49.0,$$

$$\frac{d\ln\Lambda^{(5)}}{d\ln f}=\frac{21}{23}\frac{d\ln\Lambda^{(6)}}{d\ln f}+\frac{2}{23}\frac{d\ln m_t}{d\ln f},\qquad \frac{d\ln\Lambda^{(4)}}{d\ln f}=\frac{21}{25}\frac{d\ln\Lambda^{(6)}}{d\ln f}+\frac{2}{25}\Big(\frac{d\ln m_t}{d\ln f}+\frac{d\ln m_b}{d\ln f}\Big),$$

with the quark sensitivities $d\ln m/d\ln f=-1+\frac12[d\ln\zeta(c_L)^2/d\ln c_L+d\ln\zeta(-c_R)^2/d\ln(-c_R)]$ (from $m=e/(fL)$ with the first-order root $e\simeq\zeta(c_L)\zeta(-c_R)$, $\zeta(c)^2=c\,e^{-c}/\sinh c$, of section 6.2 and localisations $c=f\mu L$ proportional to f; DERIVED), $d\ln m_q/d\ln f=-18.82$, -18.01, -14.90, -12.15, -10.89, -6.77 for u, d, s, c, b, t,

$$\frac{d\ln m_p}{d\ln f}=\Big(1-\frac{\sigma_{\pi N}+\sigma_s}{m_p}\Big)\frac{d\ln\Lambda^{(3)}}{d\ln f}+\frac{\sigma_{\pi N}}{m_p}\,\frac12\Big(\frac{d\ln m_u}{d\ln f}+\frac{d\ln m_d}{d\ln f}\Big)+\frac{\sigma_s}{m_p}\frac{d\ln m_s}{d\ln f}=-46.1$$

($\sigma_{\pi N}=45$ MeV, $\sigma_s=40$ MeV, from memory; one loop, ESTIMATE): the nucleon mass is the most block-sensitive mass of UM-2. The result: $\delta\ln f\le1.1\times10^{-34}$, at the freeze-out, where the hot plasma has $\bar p=3.1\times10^{12}$ GeV^4

(mostly the QCD interaction measure with $d\ln\Lambda/d\ln f=-60$) and $\rho-3p_3=2.2\times10^{11}$ GeV^4; at most 9×10^{-49} from nucleosynthesis to today; the gain of the loop mass → source → block → mass is at most 1.1×10^{-33}, so the iteration converges at once. (Version 3 as first written kept only the hidden pressure of cold matter and only the $\delta\bar p$ term of the response and found 5.6×10^{-45}; the conclusion, a negligible back-reaction, is unchanged.)

Domain of validity. The I12 benchmark changes the strong coupling by 10 % through the thermal dark-matter density only at $T=27$ TeV ($M/T=1.6$): the coupled effective description holds through the freeze-out and everything later.

10 Three neutrino flavours: masses, mixing and flavour conversion (DERIVED)

10.1 Masses and mixing

The neutrinos are Dirac fermions whose masses are x_0 eigenvalues (section 6.2: the lepton doublet zero mode $c_L=5.9529$ mixed at the tip with the singlet zero modes $c_R=-35.0452$, -32.8481, -31.0702). The lightest mass is not measured; UM-2 takes (ASSUMED) the normal ordering, the best fit of NuFIT 6.0 (the inverted ordering is disfavoured by $\Delta\chi^2=6.1$), and

$$m_1=1.000\ \text{meV},\qquad m_2=\sqrt{m_1^2+\Delta m^2_{21}}=8.712\ \text{meV},\qquad m_3=\sqrt{m_1^2+\Delta m^2_{31}}=50.140\ \text{meV},$$

with $\Delta m^2_{21}=7.49\times10^{-5}$ eV^2 and $\Delta m^2_{31}=2.513\times10^{-3}$ eV^2 (NuFIT 6.0). The sum $\sum m_\nu=0.0599$ eV lies above the minimum 0.0588 eV of the normal ordering and below the DESI DR2 bound 0.064 eV (95 %, ΛCDM), which allows $m_1<4.1$ meV. The mixing is the neutrino tip matrix, $U=\kappa_\nu$, with the NuFIT 6.0 angles and phase of section 6.2. A neutrino produced by the charged current with the charged lepton ℓ_α is the flavour state $|\nu_\alpha\rangle=\sum_i U^*_{\alpha i}|\nu_i\rangle$ ($\alpha=e,\mu,\tau$).

10.2 The flavour evolution

At energies far below m_W the charged and neutral currents (catalogue rows 7 and 8) act as the four-fermion interaction with $G_F/\sqrt2=g^2/(8m_W^2)$ (STANDARD; UM-2 uses the measured $G_F=1.1663788\times10^{-5}$ GeV^{-2}). Coherent forward scattering in matter then adds potentials to the free propagation, and the flavour amplitudes ψ_α of a neutrino of energy E evolve as (STANDARD)

$$i\frac{d\psi}{dx_4}=\mathcal{H}\psi,\qquad \mathcal{H}=\frac{1}{2E}\,U\,\mathrm{diag}(m_1^2,m_2^2,m_3^2)\,U^\dagger+\mathrm{diag}(V_{CC},0,0)+V_{NC}\,1,$$

$$V_{CC}=\sqrt2\,G_F n_e,\qquad V_{NC}=-\frac{G_F n_n}{\sqrt2}$$

(n_e, n_n the electron and neutron densities; V_{NC} is flavour universal and drops out of the probabilities); for antineutrinos $U\to U^*$ and $V_{CC},V_{NC}\to -V_{CC},-V_{NC}$. In vacuum

$$P(\nu_\alpha\to\nu_\beta)=\Big|\sum_i U_{\beta i}U^*_{\alpha i}\,e^{-im_i^2L/(2E)}\Big|^2,\qquad P(\nu_\alpha\to\nu_\beta)=\bar P(\nu_\beta\to\nu_\alpha)\ \text{(CPT)},$$

$$P(\nu_\mu\to\nu_e)-P(\nu_e\to\nu_\mu)=-16J\sin\Delta_{21}\sin\Delta_{31}\sin\Delta_{32},\qquad \Delta_{ij}=\frac{\Delta m^2_{ij}L}{4E}.$$

In a dense neutrino gas (the early Universe, a supernova core) the state of the neutrinos of momentum $\vec p$ is a 3×3 density matrix $\rho_{\vec p}$ (and $\bar\rho_{\vec p}$ for antineutrinos), which obeys the quantum kinetic equation (STANDARD)

$$i(\partial_{x_4}+\vec v_p\cdot\nabla)\rho_{\vec p}=\big[\Omega_{\vec p}+\mathrm{diag}(V_{CC},0,0)+V_{\nu\nu},\rho_{\vec p}\big]+i\,\mathcal{C}[\rho],\qquad \Omega_{\vec p}=\frac{U\,\mathrm{diag}(m_i^2)\,U^\dagger}{2p},$$

$$V_{\nu\nu}=\sqrt2\,G_F\int\frac{d^3q}{(2\pi)^3}\,(1-\vec v_p\cdot\vec v_q)\big(\rho_{\vec q}-\bar\rho_{\vec q}\big),$$

with $\vec v_p=\vec p/p$ and $\mathcal{C}$ the collision term.

10.3 Vacuum, matter, the Sun and accelerators

The computed evolution (exact matrix exponentials, the layer products for varying densities) reproduces the vacuum formula to 3.9×10^{-15}, unitarity, CPT and the T-odd difference (DERIVED). Representative results (DERIVED; normal ordering, $\delta = 212°$):

- reactor $\bar{\nu}_e$ of 4 MeV survive with $\bar{P}_{ee} = 0.918$ at 1.65 km, 0.226 at 52.5 km and 0.267 at 180 km; atmospheric ν_μ have their first survival minimum, 0.006, at $L/E = 1516$ km/GeV;
- the MSW resonance of the 1-3 sector, $2EV_{CC} = \Delta m^2_{ee} \cos 2\theta_{13}$ with $\Delta m^2_{ee} = c^2_{12}\Delta m^2_{31} + s^2_{12}\Delta m^2_{32}$, lies at 11.135 GeV in rock of 2.8 g/cm^3 ($Y_e = 0.5$; the exact three-flavour resonance $|U^m_{e3}|^2 = 1/2$ is at 11.133 GeV); that of the 1-2 sector at the centre of the Sun, $2EV_{CC}c^2_{13} = \Delta m^2_{21} \cos 2\theta_{12}$, at 1.93 MeV;
- $\nu_\mu \to \nu_e$ appearance at the long-baseline experiments (matter of constant density):

experiment	L	E	ρ (g/cm^3)	$P(\nu_\mu \to \nu_e)$	$\bar{P}(\bar{\nu}_\mu \to \bar{\nu}_e)$	vacuum $P(\nu_\mu \to \nu_e)$
T2K	295 km	0.6 GeV	2.6	0.0517	0.0313	0.0478
NOvA	810 km	2.0 GeV	2.84	0.0481	0.0236	0.0404
DUNE	1300 km	2.5 GeV	2.848	0.0701	0.0195	0.0491

- solar ν_e arrive at the Earth as an incoherent mixture of mass states, with the adiabatic survival probability

$$P_{ee} = c^4_{13}\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta^m_{12}\Big) + s^4_{13}, \qquad \cos 2\theta^m_{12} = \frac{\cos 2\theta_{12} - A}{\sqrt{(\cos 2\theta_{12} - A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}c^2_{13}}{\Delta m^2_{21}},$$

  evaluated at the production point in the electron density $n_e = N_A \min(245\, e^{-10.54\, r/R_\odot}, 100)$ cm^{-3} (ASSUMED profile): $P_{ee} = 0.541$ (pp, 0.3 MeV), 0.520 (^{7}Be), 0.498 (pep), 0.377, 0.319, 0.306 (^{8}B at 5, 10, 15 MeV); the numerical three-flavour evolution through the solar density agrees with the adiabatic formula to 1.3×10^{-3}: the conversion is vacuum dominated at low energy and matter dominated (MSW) for ^{8}B.

10.4 The early Universe and N_{eff}

The $e^\pm$ plasma gives the electron neutrinos a thermal potential (STANDARD)

$$V_e - V_\mu = -\frac{8\sqrt{2}\, G_F p}{3m^2_W}\, \rho_{e^\pm}, \qquad \rho_{e^\pm} = \frac{7\pi^2}{60}T^4,$$

which suppresses the mixing until it falls below $\Delta m^2 \cos 2\theta/(2p)$; at the mean momentum $p = 3.15T$ the atmospheric oscillations start at $T = 4.99$ MeV and the solar ones at $T = 2.39$ MeV (DERIVED): flavour conversion is active while the neutrinos decouple. Decoupling is computed from the temperature equations (STANDARD form, Escudero 2020; MeV units)

$$\frac{dT_\gamma}{dx_4} = -\frac{4H\rho_\gamma + 3H(\rho_e + p_e) + 3HT_\gamma \frac{dP_{int}}{dT_\gamma} + \frac{\delta\rho_{\nu_e}}{\delta x_4} + 2\frac{\delta\rho_{\nu_\mu}}{\delta x_4}}{\frac{d\rho_\gamma}{dT_\gamma} + \frac{d\rho_e}{dT_\gamma} + T_\gamma \frac{d^2P_{int}}{dT^2_\gamma}}, \qquad \frac{dT_{\nu_\alpha}}{dx_4} = -HT_{\nu_\alpha} + \frac{\delta\rho_{\nu_\alpha}/\delta x_4}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta x_4} = \frac{G^2_F}{\pi^5}\Big[4(g^2_{eL} + g^2_{eR})F(T_\gamma, T_{\nu_e}) + 2F(T_{\nu_\mu}, T_{\nu_e})\Big], \qquad \frac{\delta\rho_{\nu_\mu}}{\delta x_4} = \frac{G^2_F}{\pi^5}\Big[4(g^2_{\mu L} + g^2_{\mu R})F(T_\gamma, T_{\nu_\mu}) - F(T_{\nu_\mu}, T_{\nu_e})\Big],$$

$$F(T_1, T_2) = 32 f_a(T^9_1 - T^9_2) + 56 f_s T^4_1 T^4_2 (T_1 - T_2), \qquad f_a = 0.884, \quad f_s = 0.829,$$

with $H = \sqrt{8\pi\rho/3}/M_{Pl}$, $\rho_\gamma = \pi^2 T^4_\gamma/15$, $\rho_{\nu_\alpha} = 7\pi^2 T^4_{\nu_\alpha}/120$ per flavour, the $e^\pm$ energy density and pressure with the electron mass, ν_τ following ν_μ, the effective couplings $g_{eL} = 0.727$, $g_{eR} = 0.233$, $g_{\mu L} = -0.273$, $g_{\mu R} = 0.233$ (or the UM-2 tree-level values $\frac{1}{2} + s^2$, s^2, $-\frac{1}{2} + s^2$, s^2 with $s^2 = 0.22363$), and the finite-temperature QED pressure of the $e^\pm\gamma$ plasma, $P_{int} = P_2 + P_3$,

$$P_2 = -\frac{e^2T^2}{12\pi^2}J_1 - \frac{e^2}{8\pi^4}J^2_1 + \frac{e^2m^2_e}{16\pi^4}J_2, \qquad P_3 = \frac{e^3T}{12\pi^4}I^{3/2}, \qquad \rho_{int} = -P_{int} + T\frac{dP_{int}}{dT},$$

$$J_1=\int_0^\infty dp\,\frac{p^2}{E}n_D,\qquad J_2=\int_0^\infty\!\!\int_0^\infty dp\,dq\,\frac{pq}{E_pE_q}\ln\Big|\frac{p+q}{p-q}\Big|n_D(p)n_D(q),$$
$$I=\int_0^\infty dp\,\frac{p^2+E^2}{E}n_D,\qquad n_D=\frac{2}{e^{E/T}+1}\qquad (E=\sqrt{p^2+m_e^2}).$$

Then

$$N_{eff}=\Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4+2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big].$$

With efficient oscillations the three flavours share one temperature, $dT_\nu/dx_4 = -HT_\nu + (\delta\rho_{\nu_e}/\delta x_4 + 2\delta\rho_{\nu_\mu}/\delta x_4)/(3\,d\rho_\nu/dT_\nu)$. The method reproduces the published values of the same method (DERIVED: instantaneous decoupling 3.0002, with the QED corrections 3.0108 and 3.0098; Fermi-Dirac energy transfer 3.0393, with QED 3.0495 and 3.0485; published 3.000, 3.011, 3.010, 3.039, 3.049). For UM-2 it gives $N_{eff}=3.0476$ with separate ν_e and $\nu_{\mu,\tau}$ temperatures (published with separate temperatures: 3.048) and 3.0485 with one common temperature; $T_{\nu_e}/T_\gamma=0.71732$, $T_{\nu_\mu}/T_\gamma=0.71621$, the comoving neutrino temperatures being raised by 0.265 % (ν_e) and 0.109 % ($\nu_{\mu,\tau}$) by e^+e^- annihilation. The plasma at MeV temperatures is that of the three-flavour Standard Model; the only extra radiation is that of the fold bands produced by freeze-in through I11, $\Delta N_{eff}=0.001$ (neither fold band is ever thermalised; section 11.2), so UM-2 adopts for the neutrinos the precision Standard-Model value with oscillations and the electron mass in the rates,

$$N_{eff}=3.0432\pm0.0002\qquad\text{(Cielo, Escudero, Mangano and Pisanti 2023).}$$

10.5 The cosmic neutrino background

After decoupling the flavour states lose their coherence: the CNB of UM-2 is an incoherent mixture of the three MASS eigenstates with the temperature and densities (DERIVED; instantaneous decoupling; $T_{CMB}=2.7255$ K, STANDARD)

$$T_\nu=\Big(\frac{4}{11}\Big)^{1/3}T_{CMB}=1.9454\text{ K},\qquad n_i=\frac{3\zeta(3)}{4\pi^2}\,2T_\nu^3=112.0\text{ cm}^{-3}\text{ per mass state }(\nu+\bar\nu),\qquad 336\text{ cm}^{-3}\text{ in all,}$$

$$\Omega_\nu h^2=\frac{1}{\rho_c/h^2}\sum_i 2\int\frac{d^3p}{(2\pi)^3}\frac{\sqrt{p^2+m_i^2}}{e^{p/T_\nu}+1}=6.382\times10^{-4}\qquad(\rho_c/h^2=1.05368\times10^4\text{ eV/cm}^3)$$

(6.451×10^{-4} with the heated spectrum of $N_{eff}=3.0432$: the degeneracy $N_{eff}/3$ of each state corresponds to a Fermi-Dirac spectrum of temperature $(N_{eff}/3)^{1/4}T_\nu$, so the non-relativistic density is multiplied by $(N_{eff}/3)^{3/4}$). The mean velocities today are $\langle v\rangle=\int_0^\infty p^2(p/E_i)f\,dp/\int_0^\infty p^2 f\,dp$ with $f=1/(e^{p/T_\nu}+1)$, $E_i=\sqrt{p^2+m_i^2}$:

state	mass	mean velocity today	ν_e content $\lvert U_{ei}\rvert^2$	ν_μ content	ν_τ content
ν_1	1.000 meV	$0.435\,c$	0.677	0.112	0.211
ν_2	8.712 meV	$0.0604\,c$	0.301	0.428	0.271
ν_3	50.140 meV	$0.0105\,c$	0.022	0.460	0.518

10.6 Supernovae and the dense neutrino gas

MSW in the envelope. In the envelope, with the profile $\rho=10^4\,(10^9\,\text{cm}/r)^3$ g/cm^3 and $Y_e=0.5$ (ASSUMED), a resonance lies at $n_e=\Delta m^2\cos2\theta/(2\sqrt2\,G_FE)$; it is crossed adiabatically when the adiabaticity

$$\gamma=\frac{\Delta m^2\sin^2 2\theta}{2E\cos2\theta\,|d\ln n_e/dr|}\gg1,\qquad P_{LZ}=e^{-\pi\gamma/2}$$

being the Landau-Zener probability of a jump (two-flavour conditions: Δm^2_{31} and θ_{13} for the H resonance, Δm^2_{21} and θ_{12} for the L resonance, without the factor c^2_{13} of the solar condition). For 20 MeV the 1-3 (H) resonance lies at 1573 g/cm^3 ($r=1.85\times10^4$ km) with $\gamma=178$ ($P_{LZ}=2.7\times10^{-122}$): adiabatic. For the normal ordering a ν_e from the core leaves the star as ν_3 and an $\bar\nu_e$ as $\bar\nu_1$: $P(\nu_e\to\nu_e)=|U_{e3}|^2=0.022$, $\bar P(\bar\nu_e\to\bar\nu_e)=|U_{e1}|^2=0.677$ (without

collective effects and Earth matter; DERIVED). For neutrinospheres with luminosities $L = 6,\ 6,\ 4\times10^{52}$ erg/s and mean energies $\langle E\rangle = 11,\ 14,\ 15.5$ MeV for ν_e, $\bar\nu_e$, ν_x (ASSUMED), number fluxes $N \propto L/\langle E\rangle$, the mean energies at the Earth are $\langle E\rangle_{\nu_e} = [PN_e\langle E\rangle_e + (1-P)N_x\langle E\rangle_x]/[PN_e + (1-P)N_x] = 15.29$ MeV and, with $\bar P$, 14.34 MeV for $\bar\nu_e$.

The fast flavour instability. Near the neutrinosphere the neutrino gas is dense enough for $V_{\nu\nu}$ to dominate. For an axially symmetric gas with the electron-lepton-number (ELN) angular distribution $G(u) = \sqrt{2}\,G_F[n_{\nu_e}g_{\nu_e}(u) - n_{\bar\nu_e}g_{\bar\nu_e}(u)]$ ($u = \cos\vartheta$, $\int_{-1}^{1} g\,du/2 = 1$), the linearised equations for the flavour coherence S_u give plane-wave modes $e^{-i(\Omega x_4 - Kx_3)}$ along the symmetry axis x_3 with $\omega = \Omega - \Lambda^0$, $k = K - \Lambda^3$, $\Lambda^\mu = \int du/2\,G\,v^\mu$, $v = (1,u)$, and the dispersion relation (STANDARD; $\eta^{\mu\nu} = \mathrm{diag}(1,-1)$ on (x_4,x_3) in this analysis)

$$\det\Pi = 0, \qquad \Pi^{\mu\nu} = \eta^{\mu\nu} + \int_{-1}^{1}\frac{du}{2}\,G(u)\,\frac{v^\mu v^\nu}{\omega - ku},$$

whose axisymmetric branch is $(1+J_0)(-1+J_2) - J_1^2 = 0$, $J_n = \int du/2\,G\,u^n/(\omega-ku)$, and whose azimuthal branch is $-1 + \int du/2\,G(1-u^2)/(2(\omega-ku)) = 0$; $\mathrm{Im}\,\omega > 0$ is an instability. In units of $\mu = \sqrt{2}G_F n_{\nu_e}$ (DERIVED): with an ELN crossing (forward-peaked antineutrinos, $\alpha = n_{\bar\nu_e}/n_{\nu_e} = 0.8$) the largest growth rate is $0.088\,\mu$ (homogeneous mode $0.018\,\mu$); with linear distributions ($g_{\nu_e} = 1 + 0.2u$, $g_{\bar\nu_e} = 1 + 0.6u$, $\alpha = 0.9$) it is $0.098\,\mu$ although the homogeneous mode is stable; without a crossing no mode grows, as the general theorem (Morinaga 2022) states. At $n_{\nu_e} = 10^{32}$ to 10^{33} cm^{-3} the instability grows by e in 6×10^{-10} to 6×10^{-11} s over 18 to 1.8 cm, a million times faster than the hydrodynamics.

Flavour conversion and the explosions. Conversion moves energy from the electron-flavour neutrinos, which heat the matter behind the shock through charged-current absorption, to the heavy-lepton neutrinos, which do not; a schematic estimate of the heating, $q \propto N_e\langle E^2\rangle_e + N_{\bar e}\langle E^2\rangle_{\bar e}$ with $\langle E^2\rangle = \frac{5}{4}\langle E\rangle^2$ (Fermi-Dirac-like spectra with the pinching parameter 3), compares full flavour equipartition, $q' = \frac{1}{3}(N_e\langle E^2\rangle_e + 2N_x\langle E^2\rangle_x) + \frac{1}{3}(N_{\bar e}\langle E^2\rangle_{\bar e} + 2N_x\langle E^2\rangle_x)$, and ELN-conserving pair conversion of $\Delta = \frac{2}{3}(N_{\bar e} - N_x)$ pairs, $q'' = (1-\Delta/N_e)N_e\langle E^2\rangle_e + (1-\Delta/N_{\bar e})N_{\bar e}\langle E^2\rangle_{\bar e}$, with q: the factors 0.884 and 0.760 for the neutrinospheres above (DERIVED, schematic). In 195 collapse simulations (9 to 120 solar masses) with instantaneous flavour equipartition below a baryon density ρ_c, Gogilashvili and Tamborra (2026) find the fraction of failed explosions 25.6 % without conversion and 50.8, 77.4, 93.3, 96.4 % for $\rho_c = 10^9$, 10^{10}, 10^{11}, 10^{13} g/cm^3 (27.0 % and 47.6, 61.9, 73.9, 88.3 % weighted with the Salpeter mass function), most strongly between 16 and 30 solar masses, with lighter neutron stars from the successful explosions (STANDARD, cited).

10.7 The Z boson: invisible width and the widths of the three generations

With three neutrinos and no tip Z portal the invisible width counts

$$N_\nu \equiv \frac{\Gamma_{inv}(Z)}{\Gamma(Z\to\nu\bar\nu)} = 3,$$

against the measured 2.9963 ± 0.0074 (PDG 2025; 0.5 standard deviations); a fold band of Z coupling r times a neutrino's would add $2r^2$, bounded by $r^2 < 0.0056$ (section 5.7). The tree-level partial widths (STANDARD)

$$\Gamma(Z\to f\bar f) = \frac{N_c G_F m_Z^3}{6\sqrt{2}\pi}\left(g_V^2 + g_A^2\right), \qquad g_V = T_3 - 2Q\sin^2\theta_W, \quad g_A = T_3,$$

with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$ give (DERIVED) 165.9 MeV per neutrino (497.7 MeV invisible; measured 499.1 MeV), 83.4 MeV per charged lepton, 285.4 MeV for each of u, c and 367.9 MeV for each of d, s, b (the top quark is too heavy), a hadronic width of 1674.4 MeV (1737.3 MeV with the QCD factor $1 + \alpha_s/\pi$, $\alpha_s = 0.1180$) and a total of 2422.3 MeV (2485.2 MeV with the QCD factor) against the measured 2495.5 MeV; $R_\ell = \Gamma_{had}/\Gamma_{\ell\ell} = 20.83$ against 20.77. The remaining differences are the size of the electroweak loop corrections, which UM-2 does not compute (the tree-level angle of the couplings of the mass fit, $\sin^2\theta_W = 0.22363$, changes the total by +0.82 %).

10.8 Dirac or Majorana (DERIVED, version 3)

UM-2 gives the neutrinos Dirac masses: the overlap at the tip of the lepton-doublet zero mode with three sharply localised right-handed singlets ($|c_R| = 31$ to 35) with tip couplings of order one. The alternative without

right-handed neutrinos is a Majorana mass from the gauge-invariant tip operator

$$S_M = -\frac{1}{2}\int_{y=-L} d^7x\,\sqrt{|g_{tip}|}\,\sum_{i,j} m_{ij}\,\overline{N_i^c}\,N_j + \text{h.c.}, \qquad N_i = (\Sigma^\dagger L_i)_1,$$

N^c the charge conjugate; the upper component of $\Sigma^\dagger L_i$ is neutral, since $\Sigma \to U_L \Sigma U_Y^\dagger$ absorbs both the $SU(2)_L$ and the hypercharge transformations of L_i (the tip form of the Weinberg operator). On the zero modes it gives, to first order in the tip value of the doublet zero mode,

$$m_\nu = \kappa_M\,\frac{\zeta(c_L)^2}{fL}, \qquad \zeta(c_L)^2 = 8.04\times10^{-5}, \quad \frac{1}{fL} = 22.1\ \text{TeV},$$

so the measured masses need the tip couplings $\kappa_M = 5.6\times10^{-13}$, 4.9×10^{-12}, 2.8×10^{-11}: an unexplained hierarchy, whereas the Dirac masses follow from the localisation with couplings of order one; the x_0 mechanism favours Dirac neutrinos. Observable differences: neutrinoless double-beta decay, with $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i e^{i\alpha_i}|$ between 0.84 and 4.41 meV for all Majorana phases α_i ($|U_{ei}|^2 m_i = 0.677$, 2.624, 1.111 meV; a numerical scan of the phases agrees), below the reach of present experiments; and the capture of the non-relativistic CNB, twice as frequent for Majorana as for Dirac neutrinos (both helicities active). The flavour conversion is the same in both cases (the Majorana phases cancel in the oscillation probabilities).

11 The dark sector

11.1 Dark matter: the candidate (DERIVED)

The lightest odd states of Ψ_A and of Ψ_B (the same orbital in all eight blocks), two degenerate Dirac fermions of mass

$$M_{DM} = \varepsilon_{DM} = 3.10220\,m = 44.35\ \text{TeV} \qquad (m = H = 14.30\ \text{TeV}),$$

are cold ($w = 0$ at rest), electrically neutral and Z-neutral (UM-2 has no tip operator; a P_+ tip current would not reach Ψ_B at all), and stable as long as the fold parity is exact (UM-2 has no oriented tip operator and the bulk is reflection symmetric) and N_A, N_B are conserved (both hold in the good sector; the three portals commute with B). Their hidden wave number carries 90 % of ε^2: their mass is the author's x_0 mechanism in the Heisenberg sense. Since every portal is diagonal in the halves, each species freezes out separately and must supply half of $\Omega_c h^2 = 0.120$.

11.2 The thermal history and the relic (DERIVED, version 3)

Degrees of freedom. The particle content of UM-2 at the temperature T, with its x_0 masses: photon, gluons, W^+, W^-, Z (no Higgs particle), the six quarks, three charged leptons, three neutrinos; the dark matter is not counted in g_ρ, g_s (it would add less than 0.1 at 5 TeV and nothing at its freeze-out); the right-handed neutrinos and the fold bands are never thermalised, the radion and the Kaluza-Klein modes not below 10 TeV. From the exact ideal-gas integrals

$$\rho_i = \frac{g_i}{2\pi^2}\int_0^\infty \frac{p^2 E\,dp}{e^{E/T}\pm1}, \qquad p_i = \frac{g_i}{6\pi^2}\int_0^\infty \frac{p^4\,dp}{E\,(e^{E/T}\pm1)}, \qquad s_i = \frac{\rho_i + p_i}{T},$$

$$g_\rho = \frac{30}{\pi^2 T^4}\sum_i \rho_i, \qquad g_s = \frac{45}{2\pi^2 T^3}\sum_i s_i,$$

with the QCD crossover modelled by the weight $w(T) = \frac{1}{2}[1 + \tanh((T - 155\ \text{MeV})/15\ \text{MeV})]$ of the quark-gluon plasma against a gas of the lightest hadrons (ASSUMED model; masses in GeV and internal states: π 0.1380, 3; K 0.4956, 4; η 0.5479, 1; ρ 0.7753, 9; ω 0.7827, 3; K^* 0.8917, 12; N 0.9389, 8; η' 0.9578, 1; ϕ 1.0195, 3; Λ 1.1157, 4; Σ 1.1932, 12; Δ 1.232, 32; from memory) and neutrinos that decouple at 2 MeV (afterwards $T_\nu/T = (g_{s,e\gamma}/(11/2))^{1/3}$): $g_\rho = 105.75$ above the top threshold (the Standard Model of three generations without a Higgs particle), 86.2 at 10 GeV, 76.3 at 1 GeV, 10.75 at a few MeV, and today $g_s = 43/11 = 3.9091$, $g_\rho = 3.3626$, $s_0 = 2891.3\ \text{cm}^{-3}$ ($T_0 = 2.7255$ K).

Boltzmann equation. For one Dirac species ($Y = (n_\chi + n_{\bar\chi})/s$, $x = M/T$)

$$\frac{dY}{dx} = -\frac{(1+\delta)\, s\, \langle\sigma v\rangle}{2xH}\big(Y^2 - Y_{eq}^2\big), \qquad \delta = \frac{1}{3}\frac{d\ln g_s}{d\ln T},$$

$$H = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{Pl}}, \qquad s = \frac{2\pi^2}{45} g_s T^3, \qquad Y_{eq} = \frac{4M^2 T K_2(x)}{2\pi^2 s},$$

with the exact relativistic thermal average (Gondolo and Gelmini; K_1, K_2 modified Bessel functions)

$$\langle\sigma v\rangle = \frac{1}{8M^4 T K_2(x)^2}\int_{4M^2}^{\infty}\sigma(s)\,(s-4M^2)\,\sqrt{s}\,K_1(\sqrt{s}/T)\,ds,$$

and $\Omega h^2 = M Y_\infty s_0/(\rho_c/h^2)$, $\rho_c/h^2 = 1.05368\times10^{-5}$ GeV/cm^3. Checks: for an s-wave cross-section 4.4×10^{-26} cm^3/s it gives $\Omega h^2 = 0.1275$, against 0.1306 from the freeze-out formula (STANDARD, Kolb and Turner; Dirac fermion with $g=2$, $c=1/2$, $\langle\sigma v\rangle = a + 6b/x$)

$$x_f = \ln\Big[0.038\, c(c+2)\, g\, M_{Pl} M \langle\sigma v\rangle(x_f)\big/\sqrt{g_* x_f}\Big], \qquad \Omega h^2 = \frac{2\times1.07\times10^9\ \text{GeV}^{-1}\, x_f}{\sqrt{g_*}\, M_{Pl}\,(a + 3b/x_f)}$$

($M_{Pl} = G_N^{-1/2} = 1.22091\times10^{19}$ GeV), and the late-time integral is reproduced to 1.3×10^{-8}; for I12, whose cross-section is $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2M^4v^2/\pi$ per abelian channel (v the relative velocity; in total $b = (M^4/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$, $a=0$), the thermal average exceeds the non-relativistic $6b/x$ by the factor $1 + c_1/x$ with $c_1 \approx 3$.

I12 alone (the result of version 2, recomputed). The effective operator would need $G = (15.86\ \text{TeV})^{-3}$ for $\Omega h^2 = 0.060$ per species ($x_f = 31.7$, $\langle\sigma v\rangle(x_f) = 1.89\times10^{-25}$ cm^3/s). It annihilates only from $^3P_0$ ($J=0$), and partial-wave unitarity bounds the spin-averaged cross-section of unpolarised Dirac fermions by

$$\sigma \le \frac{\pi}{4p^2} \quad (p \text{ the centre-of-mass momentum}), \qquad \sigma v \le \frac{\pi}{M^2 v}, \qquad \Big\langle\frac{\pi}{M^2 v}\Big\rangle = \frac{\sqrt{\pi x}}{M^2}.$$

In version 2 the freeze-out formula (with $a=0$, $g_* = 105.75$) gave $G = (15.5\ \text{TeV})^{-3}$ at $x_f = 30.4$ and a requirement of 2.01×10^{-25} cm^3/s against the bound 5.8×10^{-26} cm^3/s, 3.5 times. With the numerical solution of version 3 the effective cross-section crosses the bound at $\sqrt{s} = 89\,510$ GeV, i.e. at the relative velocity $v = 2\beta = 0.269$ (0.274 from the non-relativistic formula $v_c = (\pi/(GM^3))^{2/3}$); at freeze-out 96 % of the annihilation rate comes from collisions above the crossing, and 75 % of the pairs of the non-relativistic Maxwell distribution are faster. A cross-section at the $^3P_0$ limit at every energy leaves $\Omega h^2 = 0.058$ per species (at its freeze-out, $x_f = 30.8$, $\langle\sigma v\rangle = 5.3\times10^{-26}$ cm^3/s against $\sqrt{\pi x}/M^2 = 5.8\times10^{-26}$ cm^3/s at the same x), and the effective operator capped at the limit leaves 0.154, 0.108, 0.084, 0.065, 0.059 for $G^2/G^2_{relic} = 1$, 3, 10, 100, 10^4: within perturbative couplings I12 overproduces the dark matter, and only a strongly coupled $^3P_0$ channel at the unitarity limit could give the relic. The benchmark $G = \pi/(2M^3)$ alone leaves $\Omega h^2 = 8.0$ per species.

With I13 (version 3). I12 at its benchmark and I13 together give each species $\Omega h^2 = 0.060$ for $G_A = 1.278\times10^{-9}$ GeV^{-2} ($x_f = 31.2$; $\langle\sigma v\rangle(x_f) = 9.5\times10^{-26}$ cm^3/s from I13, 9.9×10^{-28} cm^3/s from I12), with the I13 cross-section capped at the $J=1$ limit $3\pi/(4p^2)$, which it reaches only above $v = 0.51$ (without the cap $G_A = 1.269\times10^{-9}$ GeV^{-2}, and the uncapped cross-section would cross the bound above $v = 0.52$, 25 % of the rate). With the cap, 22 % of the freeze-out rate comes from collisions at the bound and 78 % from collisions below it, so the relic of UM-2 is a thermal freeze-out with most annihilations perturbative; the contact form of I13 is an effective description (for the quark doublet $G_f^{-1/2} = (G_A q_f \tilde I_Q^{(b)})^{-1/2} = 30$ TeV, below the energies $\sqrt{s} \approx 89$ TeV of the annihilations), whose completion by a mediator is OPEN. It assumes that the radion oscillation left over from the deflation does not dominate and dilute the Universe after the freeze-out (the moduli problem, section 14).

The fold bands and the right-handed neutrinos. I11 produces the fold bands by freeze-in (section 5.7): their energy density obeys $d(\rho_{fold}a^4)/dx_4 = 8T\gamma\, a^4$ with $a \propto g_s^{-1/3}/T$, which gives $\Delta N_{eff} = \rho_{fold}/\rho_\nu$ per flavour ($\rho_\nu = \frac{7}{8}\frac{\pi^2}{15}T_\nu^4$) $= 8.0\times10^{-6}$, 1.25×10^{-4}, 1.0×10^{-3}, 8.0×10^{-3} for $T_{max} = 2$, 5, 10, 20 TeV at the benchmark; their production rate per quantum is 0.012 of the Hubble rate at $T_{max} = 10$ TeV, so they never reach equilibrium. I12 and I13 do not act on the fold bands. The right-handed neutrinos are never thermalised (I13 couples only to quarks; the dark matter annihilates into ν_R pairs only through I11, at 2.0×10^{-17} of the I13 rate into quarks: the ratio $G_V^2\sum_i \tilde I^2_{\nu_{Ri}}/(G_A^2\sum_f N_c(q_f\tilde I_f^{(b)})^2)$, DERIVED).

The self-interaction channel (version 3, after the review). The contact self-interaction $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ contains $-\lambda(\bar\chi_D\chi_F)(\bar\chi_F\chi_D)\,s_{DF}(y)^2$ with the transition scalar density $s_{DF}=a_Db_F+b_Da_F$ of the dark-matter orbital (a_D,b_D) and the fold zero mode $(a_F,0)$; s_{DF} is odd under the fold reflection, its square even, so the dark matter can annihilate into the fold bands. With the normalisation of I11 the four-dimensional strength is

$$G_\lambda=\frac{\lambda H\tilde I_\lambda}{2f^4v_t},\qquad \tilde I_\lambda=\int_{-L}^{0}s_{DF}^2\,e^{-3Hy}\,dy=14.38\ \text{(units of }H),$$

and the s-wave annihilation is estimated as $\sigma v=G_\lambda^2M^2/(2\pi)$ per final fold species, $N_F=2$ (the value of one vector channel; the Fierz coefficients are not computed: ESTIMATE). While the dark matter is in equilibrium ($T_{max}=10$ TeV to $x_f=31.23$) each annihilation puts $2M$ into the fold bands, $d(\rho_{fold}a^4)/dx_4=2M\,\gamma_\lambda a^4$ with $\gamma_\lambda=2n_\chi^2\,\sigma v\,N_F$ ($n_\chi=2(MT/2\pi)^{3/2}e^{-M/T}$ particles of one species, two species), which gives $\Delta N_{eff}=1.83\times10^{26}\,G_\lambda^2$ GeV^4 as long as the fold bands stay out of equilibrium. Hence $\Delta N_{eff}\le0.001$ requires $G_\lambda\le2.3\times10^{-15}$ GeV^{-2} ($1.8\times10^{-6}\,G_A$; the production rate per fold quantum is then at most 0.027 of the Hubble rate). The self-coupling $\lambda_B=7\times10^{-5}H^{-6}$ of the numerical Kohn-Sham tests would, with $v_t=H^{-3}$, give $G_\lambda=1.1\times10^{-9}$ GeV^{-2}, keep the fold bands in equilibrium down to $T=2.6$ TeV (production rate up to 6×10^9 times the Hubble rate) and add $\Delta N_{eff}=4(10.75/g_s)^{4/3}F_h=0.19$ (four Weyl species; $N_{eff}=3.23$, 2.9 standard deviations above ACT DR6). UM-2 therefore ASSUMES $G_\lambda\le2.3\times10^{-15}$ GeV^{-2}, i.e. $\lambda\le1.4\times10^{-10}H^{-6}$ for $v_t=H^{-3}$, or $v_t\ge4.9\times10^5H^{-3}$ for $\lambda=\lambda_B$ (DERIVED bound, 1 check); with such a λ the self-interaction mean fields of every Kohn-Sham state of the cosmology are negligible, and λ_B is only the value with which the Kohn-Sham machinery was tested.

11.3 Detection (DERIVED and ESTIMATE)

I13 and I11 couple the dark matter coherently to the vector currents of the quarks: per nucleon

$$G_{u,d}=G_A\,\tfrac13\,\tfrac12\big(\tilde I_Q^{(b)}+\tilde I_{u_R,d_R}^{(b)}\big)+G_V\,\tfrac13\,\tfrac12\big(\tilde I_Q+\tilde I_{u_R,d_R}\big),$$

$$f_p=2G_u+G_d,\qquad f_n=G_u+2G_d,\qquad f_N=\frac{Zf_p+(A-Z)f_n}{A},\qquad \sigma_n=\frac{\mu_N^2f_N^2}{\pi}$$

(μ_N the dark-matter-nucleon reduced mass; xenon $Z=54$, $A=131$): $f_p=1.915\times10^{-9}$, $f_n=1.917\times10^{-9}$ GeV^{-2} and $\sigma_n=4.0\times10^{-46}$ cm^2, 0.16 of the LZ limit extrapolated linearly in the mass from 2.2×10^{-48} cm^2 at 40 GeV to 44.35 TeV (2.44×10^{-45} cm^2; STANDARD, from memory; the extrapolation an ESTIMATE): allowed, and within reach of the next generation of xenon detectors. I11 alone gives 3.9×10^{-60} cm^2; I12 through the trace anomaly, $f_N=G_g\frac{8\pi}{9\alpha_s}m_Nf_{TG}$ ($f_{TG}=0.92$, $\alpha_s=0.3$), at most 6×10^{-53} cm^2 (version 2). Indirect detection: the s-wave annihilation into quark pairs continues today with $\sigma v=\sum_f N_cG_f^2M^2/(2\pi)=9.8\times10^{-26}$ cm^3/s (per pair of particle and antiparticle), the thermal value for a heavy relic. Colliders: 44 TeV is beyond present and planned machines.

11.4 Dark energy (NOT SUPPORTED)

No positive-energy state of Ψ has an equation of state $w=p_3/\rho<0$ on the static block; a $\lambda<0$ condensate tuned to $w_0=-1$ runs away ($w_a=-dw/da=-6$ at $a=1$, the coefficient of the form $w=w_0+w_a(1-a)$, a the scale factor) and its energy density turns negative at redshift $z=0.26$. The dark energy of UM-2 is the tuned residual Λ_4.

12 Cosmology: CMB, CNB, BAO

The four-dimensional effective theory is general relativity (section 8.5) with photons, the three neutrino mass eigenstates (1.000, 8.712, 50.140 meV, together $N_{eff}=3.0432$), the fold bands produced by freeze-in (massless, $\Delta N_{eff}=0.001$), so $N_{eff}=3.0442$ in all, baryons, the dark matter (the odd quanta of Ψ_A and Ψ_B) and Λ_4. Its linear perturbations (order ϵ) obey the standard Einstein-Boltzmann equations (STANDARD; Newtonian gauge, conformal time τ, primes $d/d\tau$, $\mathcal H=a'/a$, $ds^2=a^2[-(1+2\psi)d\tau^2+(1-2\phi)d\vec x^{\,2}]$, comoving wavenumber k):

$$k^2\phi+3\mathcal H(\phi'+\mathcal H\psi)=-4\pi G_Na^2\delta\rho,\qquad k^2(\phi'+\mathcal H\psi)=4\pi G_Na^2(\bar\rho+\bar P)\theta,\qquad k^2(\phi-\psi)=12\pi G_Na^2(\bar\rho+\bar P)\sigma;$$

cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$; baryons and photons coupled by Thomson scattering (n_e the free-electron density, σ_T the Thomson cross-section, $R = 4\bar\rho_\gamma/(3\bar\rho_b)$, c_s the baryon sound speed)

$$\delta_b' = -\theta_b + 3\phi', \qquad \theta_b' = -\mathcal{H}\theta_b + c_s^2 k^2\delta_b + R\,a n_e\sigma_T(\theta_\gamma - \theta_b) + k^2\psi,$$

$$\delta_\gamma' = -\tfrac{4}{3}\theta_\gamma + 4\phi', \qquad \theta_\gamma' = k^2\big(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\big) + k^2\psi + a n_e\sigma_T(\theta_b - \theta_\gamma),$$

$$F_{\gamma l}' = \frac{k}{2l+1}\big[lF_{\gamma(l-1)} - (l+1)F_{\gamma(l+1)}\big] - a n_e\sigma_T\Big[F_{\gamma l} - \frac{\delta_{l2}}{10}\big(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\big)\Big] \quad (l \geq 2),$$

$$G_{\gamma l}' = \frac{k}{2l+1}\big[lG_{\gamma(l-1)} - (l+1)G_{\gamma(l+1)}\big] + a n_e\sigma_T\Big[-G_{\gamma l} + \frac{1}{2}\big(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\big)\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big]$$

($F_{\gamma 2} = 2\sigma_\gamma$; $G_{\gamma l}$ the polarisation multipoles); and one hierarchy per neutrino mass state i (comoving momentum q, $E_i = \sqrt{q^2 + a^2 m_i^2}$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed Fermi-Dirac distribution $\bar f_\nu$)

$$\mathcal{N}_0^{(i)\prime} = -\frac{qk}{E_i}\mathcal{N}_1^{(i)} - \phi'\frac{d\ln\bar f_\nu}{d\ln q}, \qquad \mathcal{N}_1^{(i)\prime} = \frac{qk}{3E_i}\big(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}\big) - \frac{E_i k}{3q}\psi\frac{d\ln\bar f_\nu}{d\ln q},$$

$$\mathcal{N}_l^{(i)\prime} = \frac{qk}{(2l+1)E_i}\big[l\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}\big] \quad (l \geq 2).$$

They were solved with the Einstein-Boltzmann code CAMB, each neutrino mass state evolved separately with the degeneracy $N_{eff}/3$, at the Planck 2018 values (STANDARD) $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\,\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$ (the optical depth to reionisation), $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$, with θ held fixed (DERIVED): $H_0 = 67.38$ km/s/Mpc, $r_{drag} = 147.10$ Mpc, and the TT and EE spectra agree with Planck ΛCDM (one neutrino of 0.06 eV, $N_{eff} = 3.044$) to 2.5×10^{-4} and 2.7×10^{-4} ($30 \leq \ell \leq 2500$; for TT at most 0.007 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole). The BAO distances $D_M(z) = \int_0^z c\,dz'/H(z')$ (flat), $D_H = c/H(z)$ and $D_V = (zD_M^2 D_H)^{1/3}$, in units of the drag-epoch sound horizon r_d, are compared with the DESI DR2 measurements (STANDARD, arXiv:2503.14738)

tracer	z	D_V/r_d	D_M/r_d	D_H/r_d	r_{MH}
BGS	0.295	7.942 ± 0.075			
LRG1	0.510		13.588 ± 0.167	21.863 ± 0.425	-0.459
LRG2	0.706		17.351 ± 0.177	19.455 ± 0.330	-0.404
LRG3+ELG1	0.934		21.576 ± 0.152	17.641 ± 0.193	-0.416
ELG2	1.321		27.601 ± 0.318	14.176 ± 0.221	-0.434
QSO	1.484		30.512 ± 0.760	12.817 ± 0.516	-0.500
Lyα	2.330		38.988 ± 0.531	8.632 ± 0.101	-0.431

through $\chi^2 = \sum_i (d_i - d_i^{th})^T C_i^{-1}(d_i - d_i^{th})$ with $C_i = \begin{pmatrix} \sigma_M^2 & r_{MH}\sigma_M\sigma_H \\ r_{MH}\sigma_M\sigma_H & \sigma_H^2 \end{pmatrix}$ for the pairs $d_i = (D_M/r_d, D_H/r_d)$ of the anisotropic tracers and $(d - d^{th})^2/\sigma_V^2$ for BGS: $\chi^2 = 31.5$ for 13 measurements (Planck ΛCDM: 32.6) (DERIVED).

The freeze-in fold bands are free-streaming radiation; without them (the neutrino content of version 2, $N_{eff} = 3.0432$) the TT spectrum would differ by at most 1.6×10^{-4}, $H_0 = 67.37$ km/s/Mpc, the deviations from ΛCDM would be 2.8×10^{-4} (TT and EE) and the BAO $\chi^2 = 31.7$. The linear matter power spectrum gives $\sigma_8 = 0.8118$ for UM-2 (0.8119 without the freeze-in fold bands) against 0.8109 for ΛCDM (Planck 2018: 0.811, STANDARD, from memory), and $f\sigma_8(z = 0.5) = 0.4761$ against 0.4756 (f the logarithmic growth rate); the potentials of section 4.3 come from the same solution (DERIVED, 8 checks).

13 Parameters

parameter	value	status
H	14.30 TeV	DERIVED from m_{KK}, η and f_*
Lovelock weights w, Λ	$(1, 1/32, 1/1152)$, $-10H^2$	AUTHOR
$\eta = HL$	3	ASSUMED (record)
$m = 3HM$	$M = 1/3$ ($m = H$)	ASSUMED (record)
stabiliser law	$\rho_b \propto f^{-8}$, $p_0 = p_t = \rho_b = -p_3$	ASSUMED
k_*, f_*, $G(k_*)$	21.522, 0.21556, 13.808	DERIVED
m_r	48.4 TeV	DERIVED
m_{KK}	121.3 TeV	DERIVED (fit to $\rho_0 - 1$ and m_W)
β_W, β_Z; $\cos^2\theta_W$	0.011912, 0.015343; 0.776368	DERIVED (fit; $\beta_Z = \beta_W/\cos^2\theta_W$, θ_W the angle of the couplings, fixed by m_W/m_Z and ρ_0)
tip matrices κ_e, κ_u; κ_ν; κ_d	1; U_{PMNS}; V_{CKM} (unitary)	ASSUMED (the mixing values STANDARD, measured)
doublet localisations c_L, c_Q	5.9529, 3.3823	DERIVED (fit to m_τ, m_t with $c_R = -c_L$)
singlet localisations c_R	$\nu_{1,2,3}$: -35.045, -32.848, -31.070; e, μ, τ: -14.554, -8.981, -5.953; u, c, t: -15.430, -8.768, -3.382; d, s, b: -14.626, -11.516, -7.501	DERIVED (fit to the masses)
lightest neutrino mass m_1; ordering	1.000 meV; normal	ASSUMED
Δm^2_{21}, Δm^2_{31}; $\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, δ	7.49×10^{-5}, 2.513×10^{-3} eV^2; 0.308, 0.470, 0.02215, 212°	STANDARD (NuFIT 6.0)
N_{eff}	3.0432 (neutrinos) +0.0010 (freeze-in fold bands) = 3.0442	STANDARD for the neutrinos (precision value; UM-2 computes 3.0476 to 3.0485), DERIVED for the fold bands
c_Z, c'_Z (the tip Z portal of version 1)	0 (removed)	ASSUMED (bounded by N_ν, disfavoured by N_{eff})
I11 strength $G_V = c_V H/(2f^4 v_t \Lambda_\star^6)$	4.38×10^{-17} GeV^{-2} (freeze-in $\Delta N_{eff} = 0.001$ from $T_{max} = 10$ TeV)	ASSUMED benchmark (version 3; version 2: 0)
I12 strength $G = (8G_g^2 + 3G_W^2 + G_B^2)^{1/2}$	$\pi/(2M_{DM}^3) = (38.2\text{ TeV})^{-3}$; universal $c/\Lambda_\star^3 = 1.106 \times 10^{-14}$ GeV^{-3}	ASSUMED benchmark (unitarity safe; version 3)
I13 strength $G_A = c_A H/(2f^4 v_t \Lambda_\star^6)$	1.278×10^{-9} GeV^{-2}	DERIVED from $\Omega_c h^2 = 0.120$ (version 3)
T_{max} (highest temperature of the hot Universe)	10 TeV	ASSUMED
self-interaction strength $G_\lambda = \lambda H \tilde{I}_\lambda/(2f^4 v_t)$	$\leq 2.3 \times 10^{-15}$ GeV^{-2} ($\lambda \leq 1.4 \times 10^{-10} H^{-6}$ for $v_t = H^{-3}$)	ASSUMED (bound from $\Delta N_{eff} \leq 0.001$; version 3 after the review)
M_{DM}	44.35 TeV	DERIVED

14 Assumptions, open problems and conflicts with data

- **ASSUMED**: the regular tip cutoff $\eta = 3$; the good sector and compact extra times; $m = H$; the block stabiliser and its f^{-8} law; the boundary assignments of the Standard-Model fields; the tip sigma field; one doublet bulk mass per sector, common to the three generations, with $c_R = -c_L$ for τ and t; unitary tip matrices $\kappa_e = \kappa_u = 1$, $\kappa_\nu = U_{PMNS}$, $\kappa_d = V_{CKM}$; the normal ordering with $m_1 = 1$ meV; the solar density profile and the supernova envelope profile of the flavour computations; the record's particle convention for

the Krein metric.

- **INVENTED**: the portals I11, I12 and I13 (the chiral tip Z portal I9 of version 1 is removed); the benchmark strengths of I11 and I12, the bound on the self-interaction $G_\lambda \leq 2.3 \times 10^{-15}$ GeV^{-2} and the highest temperature $T_{max} = 10$ TeV are ASSUMED, the strength of I13 follows from the relic abundance.
- **OPEN**: the microscopic origin of the stabiliser; the tachyonic extra-time modes outside the good sector; the closed time-like curves implied by compact extra times; the radion oscillation left over from the deflation (ESTIMATE: its Planck-suppressed decay rate $\Gamma = C_r\, m_r^3/(4\pi \bar{M}_{Pl}^2)$, with $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor C_r, gives a lifetime of $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma \bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV ($g_* = 10.75$), below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: the cosmological moduli problem, which also conditions the dark-matter relic); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the S parameter of the Higgsless tip; anomalies of the eight-dimensional bulk theory; the cosmological constant (Λ_4 tuned to $10^{-92} H^2$); the electroweak sector beyond tree level (the angle of the couplings of the $W^\pm$, Z mass fit, $\sin^2\theta_W = 0.22363$, differs from the $\overline{MS}$ value 0.23122 used in the Z couplings by the Standard-Model loop corrections, which UM-2 does not compute; the total tree-level Z width changes by 0.82 % between the two, and the computed total, 2485.2 MeV, lies 0.4 % below the measured value); the absolute neutrino mass scale and the mass ordering (only bounded, and preferred, by data); the collective flavour evolution of supernova neutrinos beyond the linear analysis; the microscopic origin of the portals I11, I12 and I13 (I12 alone could give the relic only as a strongly coupled ${}^3P_0$ channel at the unitarity limit; version 3 adds I13, whose contact form needs a mediator at or below about 90 TeV); the Fierz coefficients of the self-interaction channel; the order-two terms of the metric in detail; the primordial spectrum (A_s, n_s, tensor modes), an input of UM-2.
- **CONFLICTS WITH DATA, inherent in the order**: there is no Higgs boson while one is observed at 125 GeV; without it the longitudinal W^+W^- scattering violates perturbative unitarity near $\sqrt{8\pi}\, v \approx 1.2$ TeV, far below the Kaluza-Klein modes that could restore it. (The one-generation conflict of version 1, a total Z width of 1.23 GeV, is removed by the three generations.)